

Design Revision 01

eProjects Work Order Number: 1637041
Construction Contract Number: N40085-25-R-2611
Buildings: CEP 200 & CEP 209
Project Title: CEP 200 & 209 Office Renovations
Date: 25 April 2025

Design Revision Purpose

1. Response to PPIs

Drawing Revisions

The following drawings, dated 4/25/2025, are added to the list of drawings and accompany this Design Revision:

SHEET NO.		NAVFAC DRAWING NO.	SHEET TITLE
NO.	SHEET		
STRUCTURAL			
8.1	SF102	12.902.313	FRAMING PLAN FOR RTU SUPPORT – BUILDING CEP 200

The following drawings are revised as of 4/25/2025, and accompany this Design Revision:

SHEET NO.		NAVFAC DRAWING NO.	SHEET TITLE
NO.	SHEET		
GENERAL			
3	GI003	12,902,238	PHASING PLAN
ARCHITECTURAL			
11	AD101	12,902,246	BUILDING CEP 200 - PARTIAL FLOOR PLAN - DEMOLITION
12	AD102	12,902,247	BUILDING CEP 209 - PARTIAL FLOOR PLAN - DEMOLITION
17	AE103	12,902,252	BUILDING CEP 200 ROOF PLAN - NEW
18	AE104	12,902,252	BUILDING CEP 200 ROOF PLAN - NEW
19	AE111	12,902,254	BUILDING CEP 200 – PARTIAL FLOOR PLAN REFLECTED CEILING PLAN - NEW
21	AE121	12,902,256	TRAILER GENERAL NOTES
22	AE122	12,902,257	TEMPORARY TRAILER FLOOR PLAN - NEW
23	AE123	12,902,258	ENLARGED TRAILER PLANS - NEW
MECHANICAL			
46	MH101	12,902,281	PARTIAL FIRST FLOOR PLAN – DUCTWORK BLDG CEP 200
49	MI601	12,902,284	MECHANICAL SCHEDULES
ELECTRICAL			
56	EL102	12,902,291	BLDG CEP 209 – PARTIAL LIGHTING PLAN – NEW WORK
57	EL501	12,902,292	LIGHTING DIAGRAMS AND DETAILS
58	EL601	12,902,293	LUMINAIRE SCHEDULE
TELECOM			
72	TD102	12,902,307	BLDG CEP-209 – PARTIAL TELECOMMUNICATIONS PLAN - DEMOLITION

Specification Revisions

The following specifications, dated 4/25/2025, are added accompany this Design Revision:

DIVISION 05 – METALS

05 52 00.00 27 – NON PENETRATING PERMANENT PERSONNEL GUARDRAIL

The following specifications, dated 4/25/2025, are revised accompany this Design Revision:

DIVISION 01 – GENERAL REQUIREMENTS

01 32 17.00 20 – COST-LOADED NETWORK ANALYSIS SCHEDULE

Noted Revisions

Drawings:

Noted Revisions are not included in sheets & specification sections that have been added or re-issued. Please make the following modifications to the drawing set:

On NAVFAC Drawing No. - 12,902.237 (GI002)

LIST OF SHEETS: In response to PPI – add sheet SF101 to drawing list of sheets.

On NAVFAC Drawing No. – 12,902,291 (EL102)

BLDG CEP-209 – PARTIAL LIGHTING PLAN – NEW WORK: Show locations of emergency lighting fixtures.

On NAVFAC Drawing No. – 12,902,292 (EL501)

Delete Lighting Schedule and Lighting Schedule Notes.

On NAVFAC Drawing No. – 12,902,293 (EL601)

Add Lighting Schedule Notes.

On NAVFAC Drawing No. – 12,902,307 (TD102)

Keynotes: Keynote 1 changed to “Disconnect existing systems furniture telecommunications connection from systems furniture. Demolish data telephone cabling back to source. Demolish associated surface raceway and conduit back to nearest upstream junction box.”

Keynote 2 changed to “Disconnect and demolish existing power pole for systems furniture and data and telephone cabling back to source.”

Keynote 3 changed to “Demolish existing surface mounted data port, data cabling, and surface mounted raceway and conduit back to source.”

Specifications:

Noted Revisions are not included in sheets & specification sections that have been added or re-issued. Please make the following modifications to the specifications:

DIVISION 07 – THERMAL AND MOISTURE PROTECTION

07 84 00 – FIRESTOPPING: In Response to PPI – remove section 07 84 00 from specification.

DIVISION 13 – SPECIAL CONSTRUCTION

13 12 00 – MODULAR BUILDINGS, SECTION 2.4: In response to PPI - Modify Floor Insulation to say: "SPRAY FOAM OR KRAFT BACKED INSULATION", Modify Finish to say "Seamless resilient floor with continuous 6" base – restroom and janitor, LVT or VCT – all other rooms."

13 12 00 – MODULAR BUILDINGS, SECTION 2.4 (EXTERIOR WALLS – MISC.): In response to PPI – modify to say: Misc: stagger panel seams- top layer gypsum is held back at mate lines possible & finished on site mud-taped-painted-LEVEL 4 interior wall finish-at factory or vinyl covered gypsum wall finish at factory & finished on site by SMM

13 12 00 – MODULAR BUILDINGS, SECTION 2.8: In response to PPI – Modify paragraph to say "Provide empty 4" conduit from Trailer Comm Room to 6" by 6" NEMA 4 J-Box mounted approximately 6 feet above finished floor of temporary trailers on outside wall of temporary building for telephone services. Provide 4" conduit with 24 strand single-mode fiber from Trailer Comm Room to 6" by 6" NEMA 4 J-box mounted approximately 6 feet above finished floor of temporary trailers on outside wall of temporary building for NMCI network. Coordinate outside plant connection with NMCI and NCTAMS."

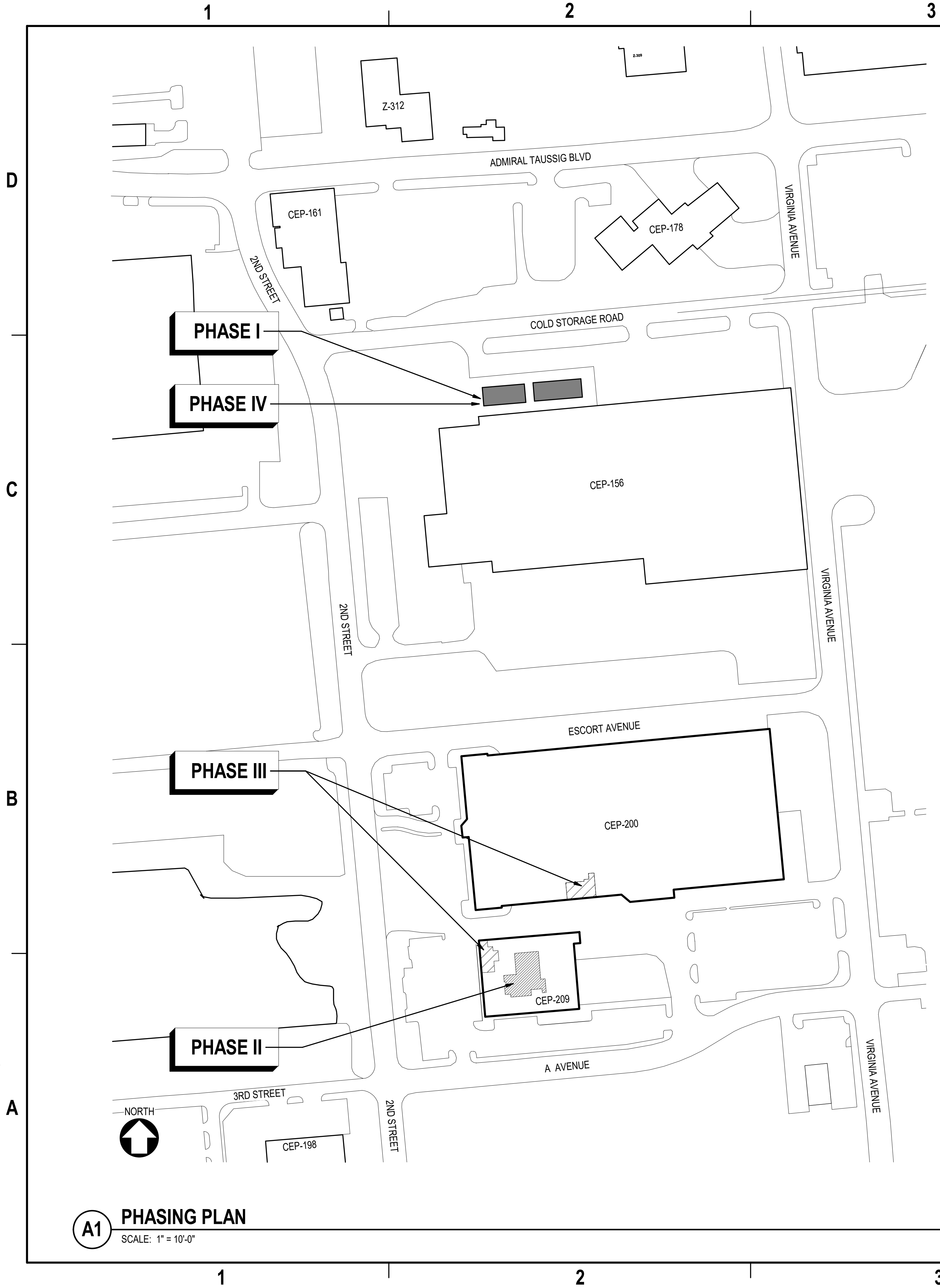
00 22 13.00 20 Supplemental Instructions to Offerors Revisions

CLIN 0002

Planned Modification Bid Notes 1.3 a – Replace reference to Section 01 30 01.00 22 DESIGN, PROCUREMENT AND INSTALLATION OF FURNITURE, FIXTURES, AND EQUIPMENT with 12 00 01.00 20 FURNITURE, FIXTURES AND EQUIPMENT (FF&E) PACKAGE.

4/29/2025 3:40:57 PM
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GI003



PHASE DESCRIPTIONS

PHASE I
TRAILER SITE IMPROVEMENTS AND
DELIVERY/INSTALLATION OF TEMPORARY TRAILERS

INSTALLATION OF 2 CONNEX BOXES FOR CUSTOMER
USE. (COMMAND TO INDICATE LOCATION)

PHASE II
CEP-209 OFFICE RENOVATION (ROOM 120)

PHASE III
CEP-200 OFFICE RENOVATION (ROOM D16) &
CEP-209 OFFICE RENOVATION (ROOM D13)

PHASE IV
REMOVAL OF TEMPORARY TRAILERS AND SITE
RESTORATION

GENERAL SHEET NOTES

- A THE BUILDING WILL REMAIN OCCUPIED AT ALL PHASES OF CONSTRUCTION. PROVIDE TEMPORARY DUSTPROOF BARRIERS BETWEEN CONSTRUCTION AREA AND ADJACENT OCCUPIED AREAS. CONDUCT SELECTIVE DEMOLITION TO MINIMIZE DISRUPTION OF THE BUILDING OPERATIONS.
- B PERFORM SELECTIVE DEMOLITION IN A MANNER AND USING EQUIPMENT THAT WILL MINIMIZE NOISE AND VIBRATIONS IN THE EXISTING BUILDING, INCLUDING BUT NOT LIMITED TO PROPER SHORING AND SEPARATION.
- C COORDINATE ANY TEMPORARY DISRUPTIONS IN SERVICE WITH CONTRACTING OFFICER. ALL UTILITY OUTAGES EFFECTING THE OCCUPIED PORTIONS OF THE EXISTING BUILDING MUST BE PLANNED AND EXECUTED AFTER NORMAL BUSINESS HOURS.
- D MAINTAIN BUILDING OCCUPANT LIFE SAFETY PER NFPA 101 AT ALL TIMES. PROVIDE CODE COMPLIANT LIFE SAFETY EGRESS PATHWAYS. THESE PATHWAYS MUST BE KEPT CLEAR AND FREE OF DEBRIS AT ALL TIMES.
- E PROVIDE TEMPORARY ENCLOSURES AS REQUIRED TO MAINTAIN THE EXISTING BUILDING IN A WEATHERTITE CONDITION. PRIOR TO BEGINNING WORK, PROTECT ALL WALLS, CEILINGS, FLOORS AND OTHER EXISTING FINISH WORK TO REMAIN.
- F UNLESS OTHERWISE NOTED, FURNITURE AND FIXTURES THAT THE COMMAND WISHES TO KEEP WILL BE REMOVED BY THE COMMAND PRIOR TO THE START OF EACH PHASE. COORDINATE SCHEDULING OF PHASING AND PREPARATION WITH THE CONTRACTING OFFICER. PERSONNEL WILL REMOVE ALL PERSONAL ITEMS AND ELECTRONICS FROM WORK STATIONS.
- G DEVELOP AND SUBMIT A DETAILED CONSTRUCTION PHASING PLAN FOR REVIEW AND APPROVAL PER SPECIFICATION 01 14 00 – WORK RESTRICTIONS. THE PLAN MUST ADDRESS HAZMAT ABATEMENT, DEMOLITION AND NEW WORK. IDENTIFY LIFE SAFETY EGRESS PATHWAYS, OCCUPANT & CONTRACTOR ENTRANCES AND EXITS, DEVELOP AND SUBMIT A "WORK AREA SCHEDULE" AS PART OF, OR AS AN APPENDIX, TO THE PHASING PLAN.
- H ALL WORK PERTINENT TO A SPECIFIC PHASE MUST BE COMPLETED PRIOR TO COMMENCING WORK ON THE NEXT PHASE.
- I AIR DISTRIBUTION SYSTEMS SERVING THE SEPARATED PHASES MUST BE KEPT SEPARATE AT ALL TIMES TO PREVENT THE MIGRATION OF HAZMAT, DUST, AND DEBRIS TO UN-RENOVATED SPACES.
- J REFER TO SHEET GI002 FOR ADDITIONAL GENERAL PROJECT NOTES.
- K REFER TO CIVIL, GEOTECHNICAL, STRUCTURAL, ARCHITECTURAL, FIRE PROTECTION, PLUMBING, MECHANICAL, ELECTRICAL, AND TELECOMMUNICATION DRAWINGS FOR ADDITIONAL PHASING INFORMATION.



APPROVED
FOR COMMANDER NAVFAC
ACTIVITY

NAVSEA				
SATISFACTORY TO DATE 5/6/24				
DES	SLG	DRW	SLG	CHK ALP
PMIDM				
BRANCH MANAGER				
CHIEF ENGINEER				
FIRE PROTECTION				

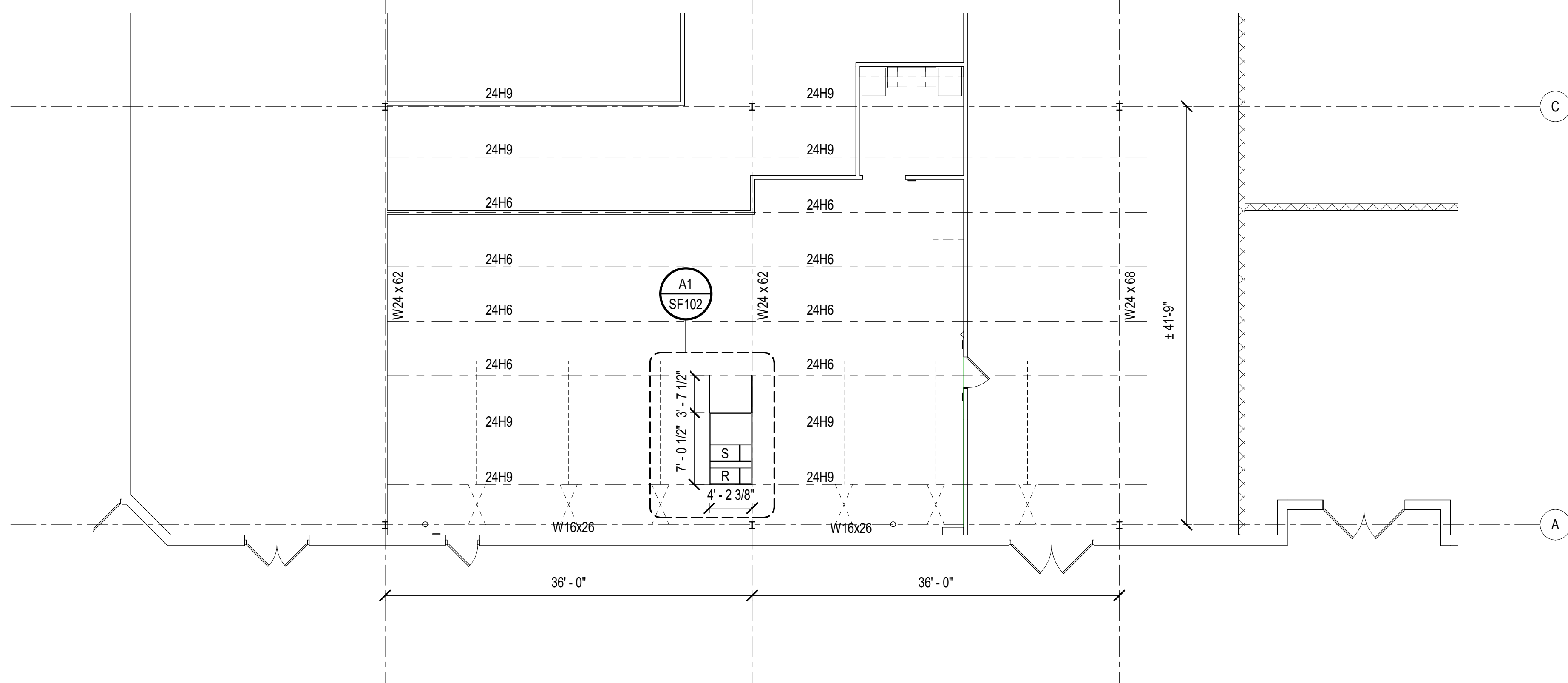
DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND – MID-ATLANTIC
MID-ATLANTIC REGIONAL OFFICE
NAVAL STATION NORFOLK
NORFOLK, VIRGINIA
BUILDINGS CEP-200 & CEP-209
FMB OFFICE RENOVATION
PHASING PLAN

SCALE: AS NOTED	
PROJECT NO.:	1637041
CONSTR. CONTR. NO.	N/A
NAVFAC DRAWING NO.	12,902,238
SHEET	3 OF 78

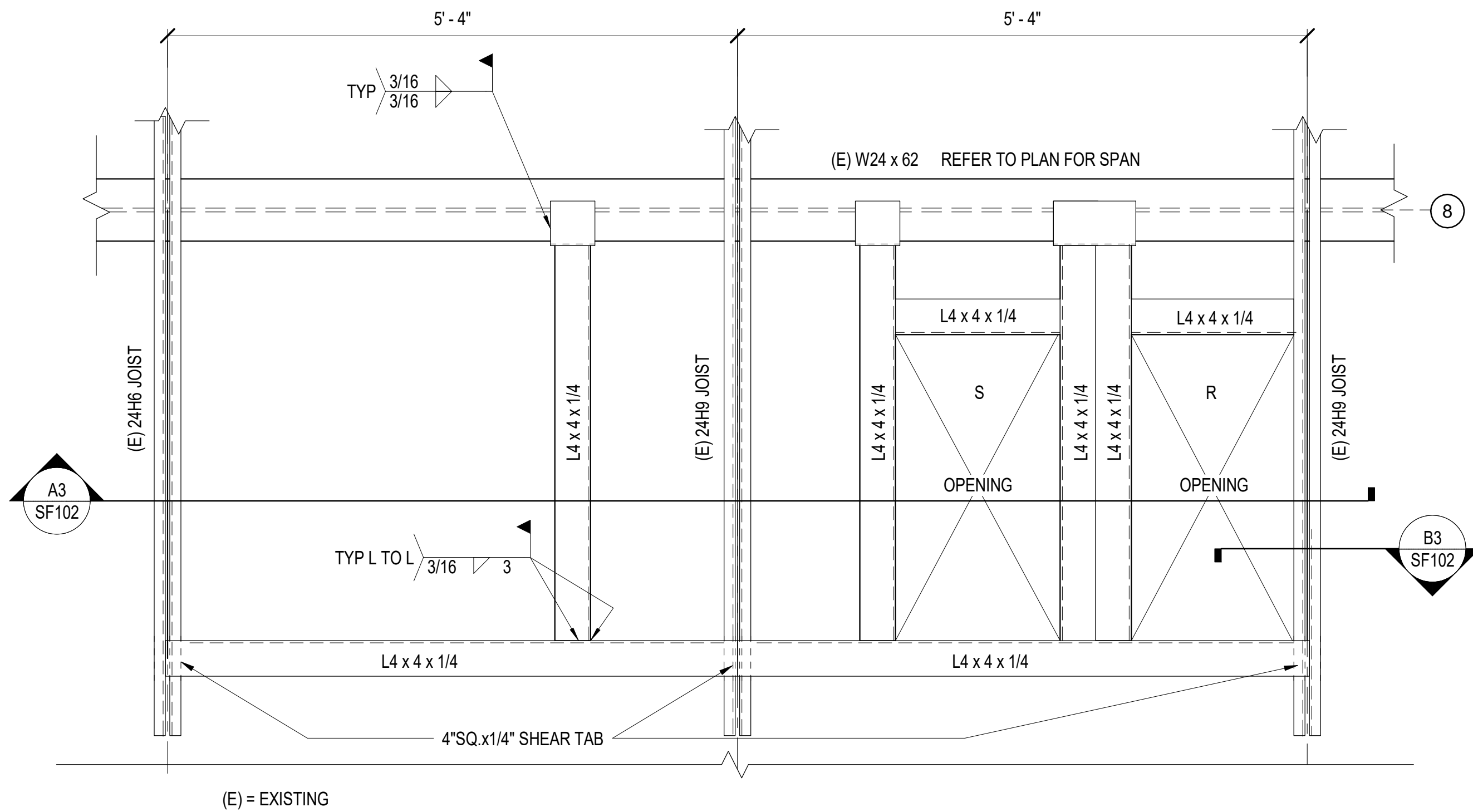
GI003

DRAWING REVISION: 25 AUGUST 2020

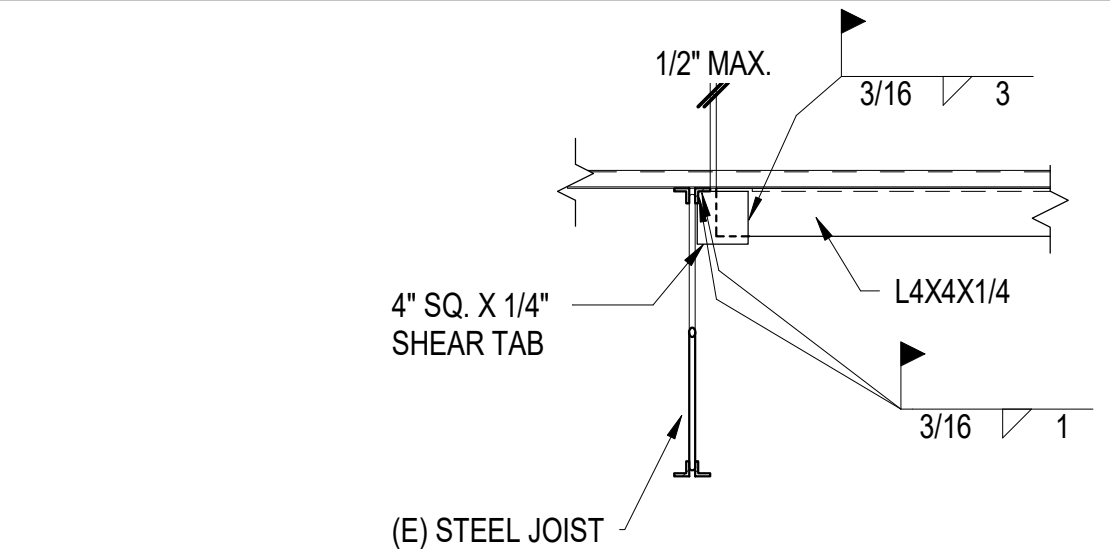
4/30/2025 2:15:05 PM SF102
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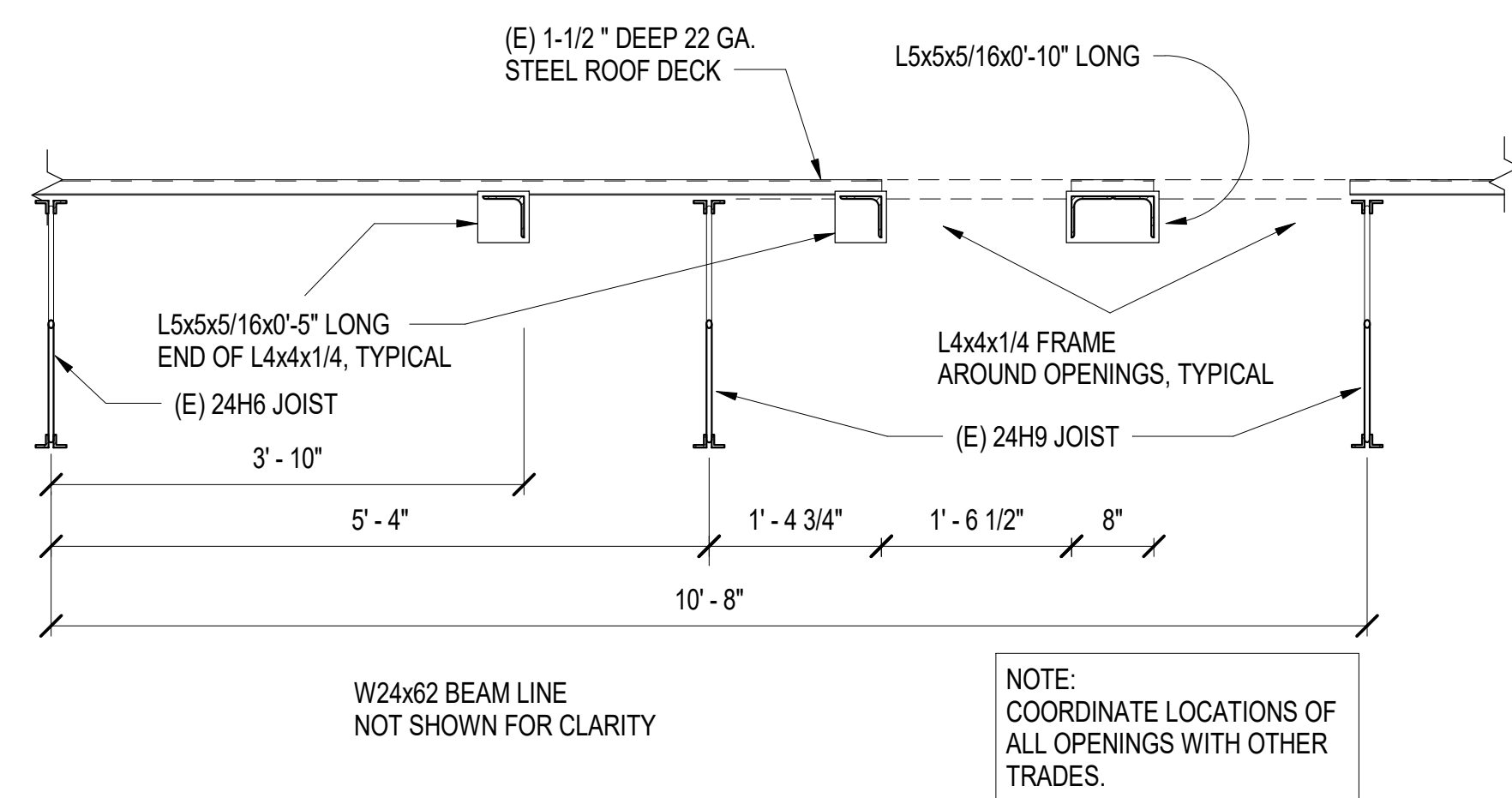
C1 ROOF FRAMING PLAN FOR ADDED ROOF TOP UNIT
SCALE: 1/8" = 1'-0"



A1 ENLARGED FRAMING AT RTU OPENING (8.A - 7.C) (ROTATED 90 DEGREE FROM PLAN)
SCALE: 1" = 1'-0"



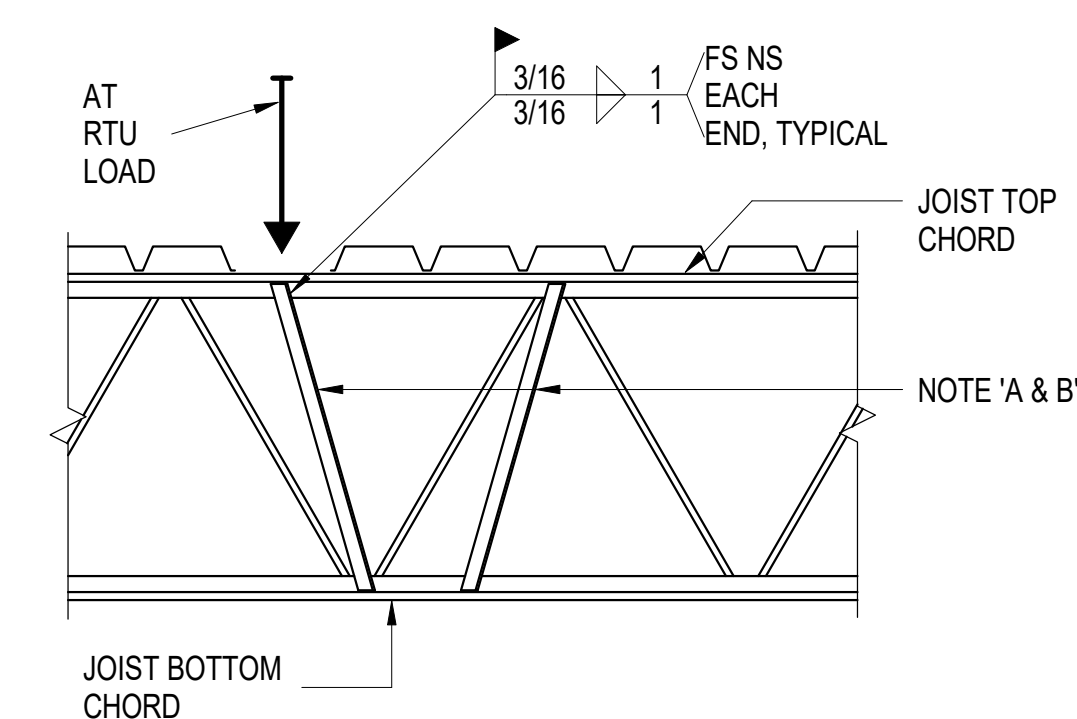
B3 ANGLE TO JOIST SHEAR TAB CONNECTION
SCALE: 3/4" = 1'-0"



A3 RTU SUPPORT FRAMING SECTION
SCALE: 3/4" = 1'-0"

GENERAL SHEET NOTES

1. NEW FRAMING MEMBERS AND DIMENSIONS SHOWN ARE BASED ON RTU TRANE WHC -060. IF DIFFERENT UNIT IS SELECTED, CONTRACTOR WILL NEED TO ADJUST FRAMING TO FIT APPROVED EQUAL RTU.
2. RTU AND CURB SHALL NOT EXCEED 1000 POUNDS.
3. CUT AND REMOVE EXISTING ROOF DECK AFTER RTU FRAMING IS INSTALLED, IMMEDIATELY PRIOR TO INSTALLING EQUIPMENT. CUT OPENING FOR DUCT PENETRATIONS ONLY.
4. COORDINATE ALL CURB AND FRAMING AROUND PACKAGED ROOF TOP UNITS WITH MANUFACTURER TO INSURE PROPER CLEARANCES FOR DUCT OPENINGS.
5. SEE JOIST LOAD STRUT DETAIL BETWEEN PANEL POINT BELOW :



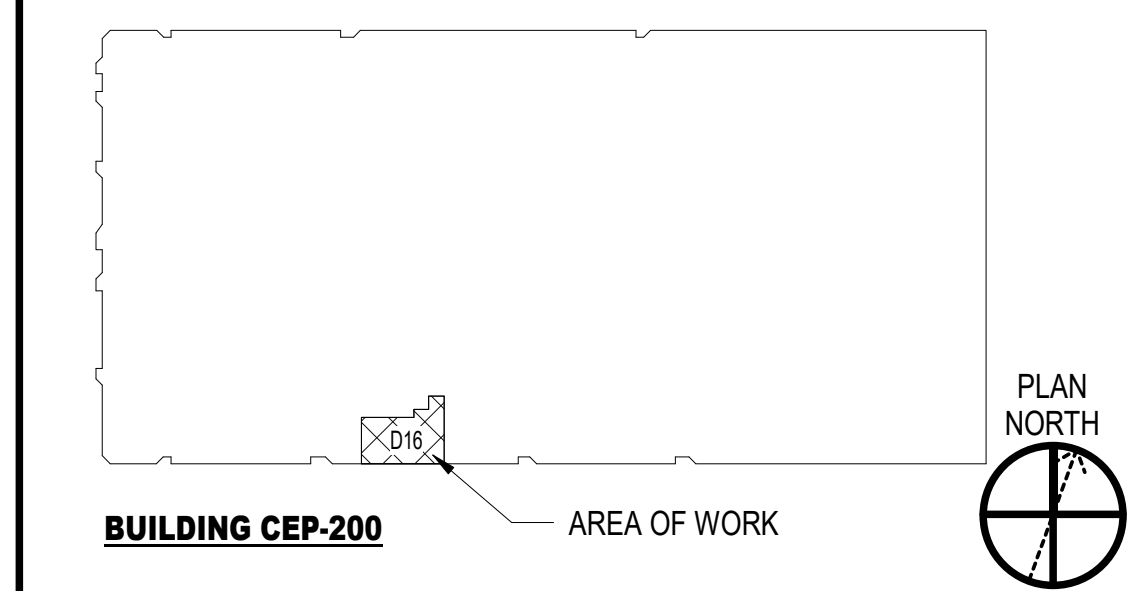
- NOTES:
- A. WHERE CONCENTRATED RTU LOADS ARE APPLIED TO TOP CHORDS BETWEEN PANEL POINTS, REINFORCE JOIST WITH DOUBLE ANGLES 2x2x1/4 AS INDICATED.
 - B. THIS DETAIL OCCURS WHERE APPLIED RTU LOADS EXCEED 50 POUNDS, BUT LIMITED TO 300 POUNDS MAXIMUM.

B5 JOIST LOAD STRUT
SCALE: 1" = 1'-0"

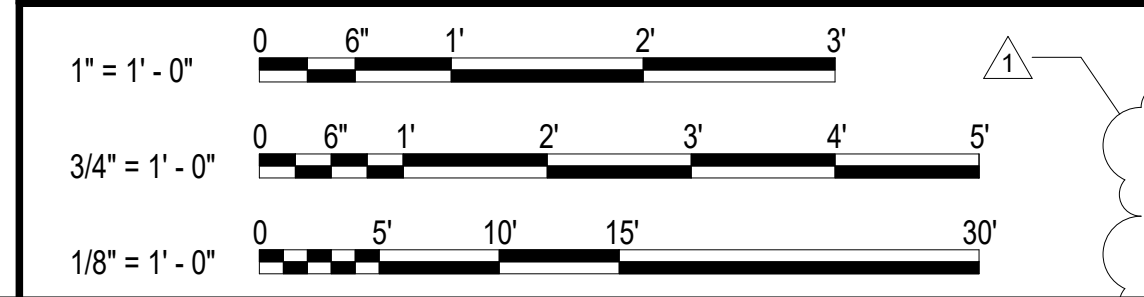
ABBREVIATIONS:

- (E) EXISTING
- TYP TYPICAL
- SQ SQUARE
- MAX MAXIMUM
- FS FAR SIDE
- NS NEAR SIDE
- RTU ROOF TOP UNIT
- S SUPPLY
- R RETURN
- L ANGLE

KEY PLAN



GRAPHIC SCALE(S)



DATE	REVISION	DESCRIPTION
4/25/25	1	MARK

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

NAVSEA

SATISFACTORY TO DATE: 5/6/24

DES: BSH | DRW: BSH | CHK: DSG

PMIDM: SLG

BRANCH MANAGER: J. M. SMITH, RA

CHIEF ENGINEER: J. MILLER, P.E.

FIRE PROTECTION

DEPARTMENT OF THE NAVY

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC

MID-ATLANTIC DISTRICT

NAVAL STATION - NORFOLK, VA

NORFOLK, VIRGINIA

BUILDINGS CEP-200 & CEP-209

FMB OFFICE RENOVATION

FRAMING PLAN FOR RTU SUPPORT

SCALE: AS NOTED

PROJECT NO.: 1637041

CONSTR. CONTR. NO.: N/A

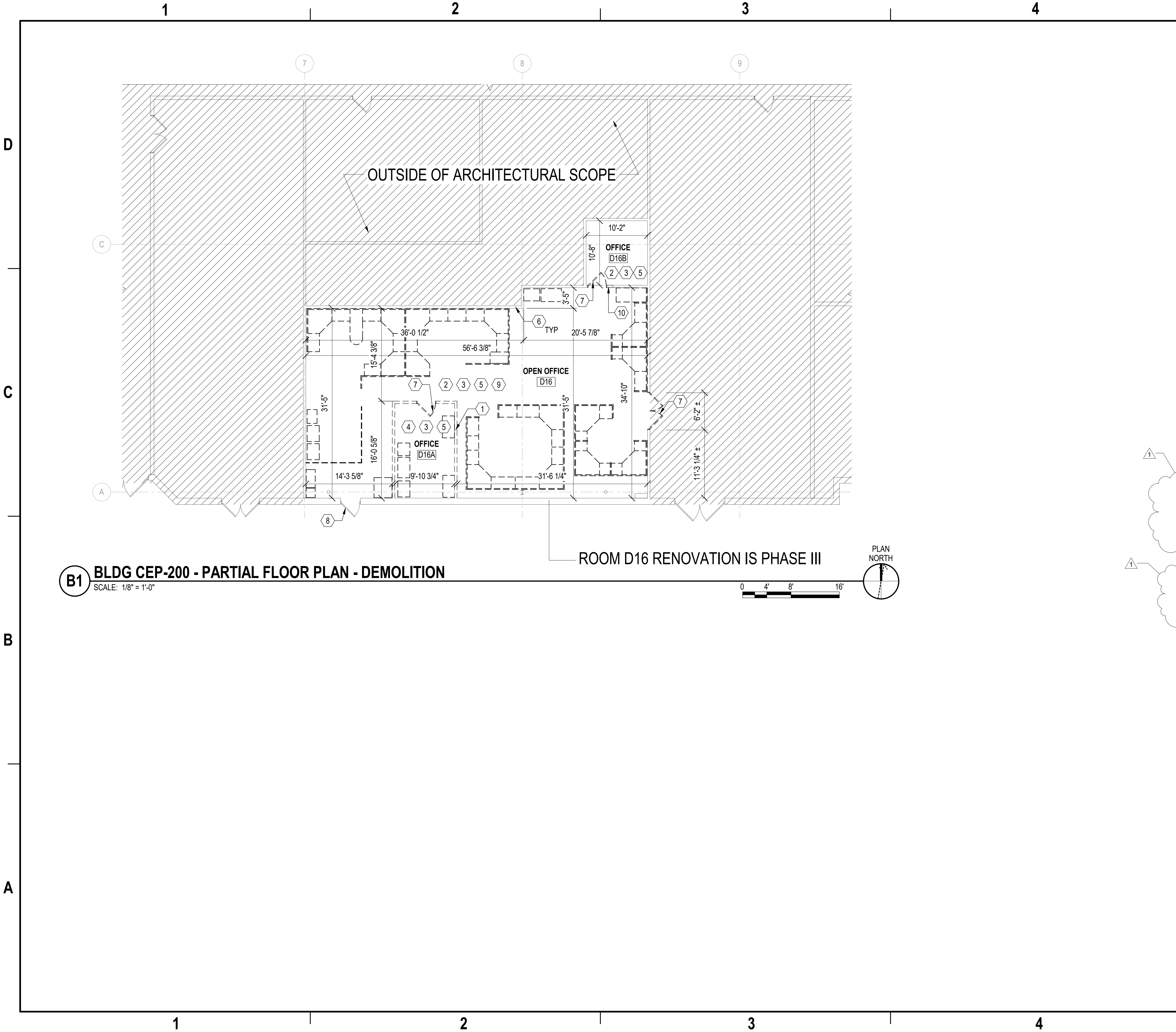
NAVFAC DRAWING NO.: 12.902.313

SHEET: 8.1 OF 78

SF102

DRAWING REVISION: 14 SEP 2023

4/29/2025 4:28:44 PM AD101
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B1 BLDG CEP-200 - PARTIAL FLOOR PLAN - DEMOLITION
SCALE: 1/8" = 1'-0"

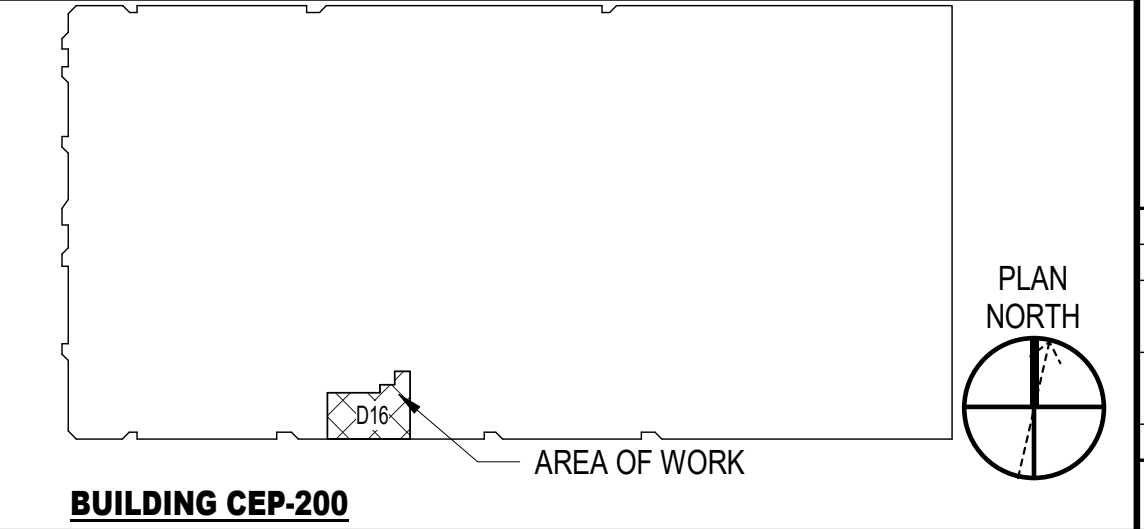
GENERAL SHEET NOTES

- A REFER TO SHEET A-001 FOR ADDITIONAL GENERAL ARCHITECTURAL NOTES
- B THE WORK INCLUDES THE DEMOLITION AND REMOVAL OF ALL CONSTRUCTION INDICATED, NOTED, OR NECESSARY TO ACCOMPLISH THE WORK AS INDICATED OR SPECIFIED. IT IS THE GENERAL INTENT OF THE DEMOLITION WORK TO COMPLETELY REMOVE ALL ITEMS OF EXISTING CONSTRUCTION, WHERE NOTED OR INDICATED IN THE DRAWINGS, INCLUDING, BUT NOT LIMITED TO: ANCHORS, FASTENERS, BRACKETS, CLIPS, RODS, HANGERS, AND SUPPORTS COMPLETE.
- C EXISTING DIMENSIONS ARE BASED ON EXISTING DRAWINGS AND FIELD WORK. VERIFY ALL DIMENSIONS AND CONDITIONS IN THE FIELD PRIOR TO PERFORMING ANY WORK. BRING ANY DISCREPANCIES TO THE ATTENTION OF THE CONTRACTING OFFICER PRIOR TO PERFORMING ANY WORK.
- D ALL INTERIOR FINISHED SURFACES WHICH ARE DAMAGED WHILE PERFORMING WORK UNDER THIS CONTRACT MUST BE PATCHED AND/OR REPLACED AND FINISHED TO MATCH ADJACENT AREAS AT NO COST TO THE GOVERNMENT.
- E DEMOLITION OF CIVIL, STRUCTURAL, FIRE PROTECTION, PLUMBING, MECHANICAL, ELECTRICAL, AND TELECOMMUNICATION ITEMS INDICATED ON THE ARCHITECTURAL DRAWINGS IS NOT INTENDED TO INCLUDE ALL NECESSARY DEMOLITION WORK FOR THESE ITEMS. REFER TO CIVIL, STRUCTURAL, FIRE PROTECTION, PLUMBING, MECHANICAL, ELECTRICAL, AND TELECOMMUNICATION DRAWINGS FOR ADDITIONAL INFORMATION.
- F DEMOLITION PLANS, INCLUDING EQUIPMENT SHOWN, IS A VISUAL REPRESENTATION OF THE CONDITION OF THE BUILDING AT THE TIME FIELD WORK WAS CONDUCTED.
- G REPAIR ALL DAMAGED CONCRETE SURFACES TO MATCH EXISTING ADJACENT CONSTRUCTION.
- H REMOVE AND SALVAGE ALL SIGNAGE AS REQUIRED FOR RENOVATION, UNLESS OTHERWISE NOTED. REINSTALL EXISTING SIGNAGE IN ORIGINAL LOCATION, UNLESS OTHERWISE NOTED. ALL DAMAGED SIGNAGE MUST BE REPLACED IN KIND.
- I REFER TO HAZARDOUS MATERIALS ABATEMENT PLAN ON HM-101.
- J REFER TO PHASING PLAN ON G1003 - OFFICE D16 TO BE RENOVATED IN PHASE III.
- K CONTRACTOR IS TO PATCH AND REPAIR ALL WALLS.

(THIS SHEET ONLY) KEYNOTES

- 1 DEMOLISH EXISTING WALL COMPLETE (+/- 52 LINEAR FEET OF WALL IN OFFICE D16)
- 2 DEMOLISH EXISTING MULTI-LAYER (CARPET AND ACT) FLOORING ALL THE WAY DOWN TO EXISTING CONCRETE SUBFLOOR. PREPARE FLOOR TO RECEIVE NEW FINISH. PROVIDE SELF LEVELING COMPOUND IF NECESSARY TO ACHIEVE A FLAT SURFACE. REFER TO SPECIFICATION 09 65 00 FOR MORE INFORMATION ABOUT REQUIRED FLOOR PREPARATION. (+/-1,800 SF TOTAL IN OFFICE D16)
- 3 DEMOLISH RESILIENT WALLBASE (+/- 309 LINEAR FEET IN RM D16)
- 4 DEMOLISH VCT FLOORING ALL THE WAY DOWN TO EXISTING SUBFLOOR. PROVIDE SELF LEVELING COMPOUND IF NECESSARY TO ACHIEVE A FLAT SURFACE. REFER TO SPECIFICATION 09 65 00 FOR MORE INFORMATION ABOUT REQUIRED FLOOR PREPARATION. (+/-155 SF TOTAL IN OFFICE D16)
- 5 ALL SYSTEMS FURNITURE IN ROOM D16 TO BE RELOCATED. CONTRACTOR IS TO DISSASSEMBLE SYSTEMS FURNITURE AND RE-ASSEMBLE IN AN ALTERNATIVE LOCATION AS INDICATED BY THE CONTRACTING OFFICER.
- 6 PATCH AND REPAIR ALL EXISTING TO REMAIN WALLS. REFER TO ELECTRICAL DEMOLITION PLAN FOR MORE INFORMATION REGARDING DEMOLITION THAT WILL EFFECT WALLS. PREPARE WALLS FOR NEW FINISH. REFER TO FINISH SCHEDULE.
- 7 DEMOLISH EXISTING DOOR AND FRAME COMPLETE.
- 8 EXISTING DOOR TO REMAIN. PROTECT DURING DEMOLITION.
- 9 ASBESTOS TESTING INDICATES ASBESTOS IS PRESENT - REFER TO ABATEMENT PLAN ON HM101
- 10 DEMOLISH PORTION OF EXISTING WALL TO ALLOW FOR NEW 4'-0" W X 6'-8" CASED OPENING. (+/- 20 SF TOTAL IN OFFICE D16)

KEY PLAN



DATE	4/29/25	APPR
SYN	DESCRIPTION	
1	REVISION 01	

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

NAVSEA

SATISFACTORY TO DATE 5/6/24

DES SLG DRW SLG CHK ALP

PMOM

BRANCH MANAGER ALP

CHIEF ENDORSE JOHN MILLER, P.E.

FIRE PROTECTION

DEPARTMENT OF THE NAVY

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND

MID-ATLANTIC

NAVAL STATION NORFOLK

NORFOLK, VIRGINIA

BUILDINGS CEP-200 & CEP-209

FMB OFFICE RENOVATION

BLDG CEP-200 - PARTIAL FLOOR PLAN - DEMOLITION

SCALE: AS NOTED

PROJECT NO.: 1637041

CONSTR. CONTR. NO. N/A

NAVFAC DRAWING NO. 12.902.246

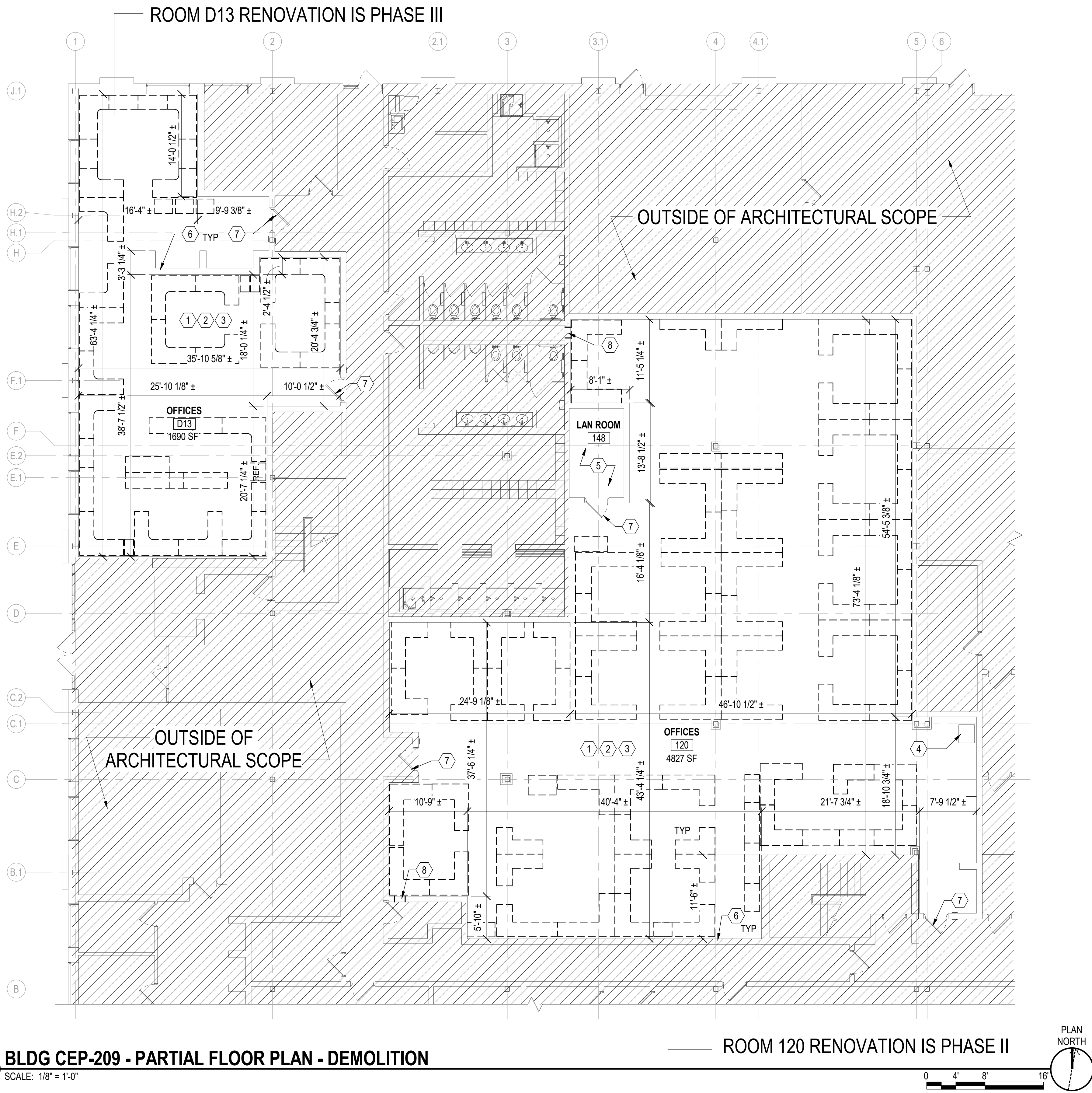
SHEET 11 OF 78

AD101

DRAWING REVISION: 25 AUGUST 2020

4/29/2025 4:29:44 PM AD102
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A1 BLDG CEP-209 - PARTIAL FLOOR PLAN - DEMOLITION
SCALE: 1/8" = 1'-0"



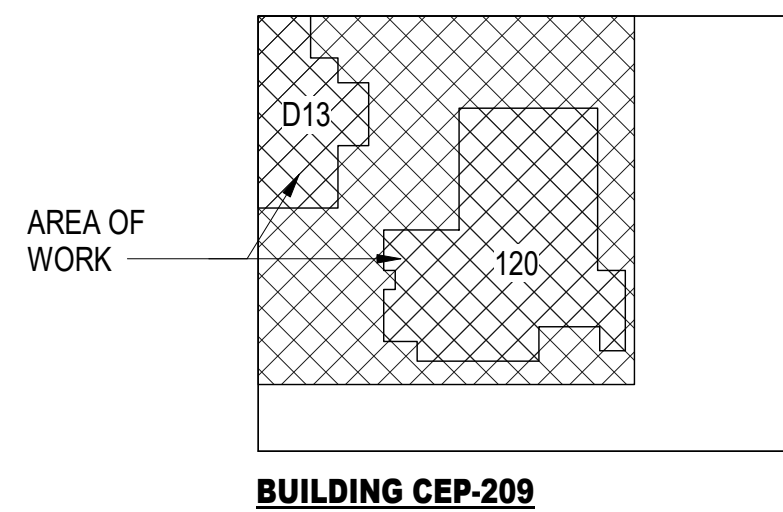
GENERAL SHEET NOTES

- A REFER TO SHEET A-001 FOR ADDITIONAL GENERAL ARCHITECTURAL NOTES
- B THE WORK INCLUDES THE DEMOLITION AND REMOVAL OF ALL CONSTRUCTION INDICATED, NOTED, OR NECESSARY TO ACCOMPLISH THE WORK AS INDICATED OR SPECIFIED. IT IS THE GENERAL INTENT OF THE DEMOLITION WORK TO COMPLETELY REMOVE ALL ITEMS OF EXISTING CONSTRUCTION, WHERE NOTED OR INDICATED IN THE DRAWINGS, INCLUDING, BUT NOT LIMITED TO: ANCHORS, FASTENERS, BRACKETS, CLIPS, RODS, HANGERS, AND SUPPORTS COMPLETE.
- C EXISTING DIMENSIONS ARE BASED ON EXISTING DRAWINGS AND FIELD WORK. VERIFY ALL DIMENSIONS AND CONDITIONS IN THE FIELD PRIOR TO PERFORMING ANY WORK. BRING ANY DISCREPANCIES TO THE ATTENTION OF THE CONTRACTING OFFICER PRIOR TO PERFORMING ANY WORK.
- D ALL INTERIOR FINISHED SURFACES WHICH ARE DAMAGED WHILE PERFORMING WORK UNDER THIS CONTRACT MUST BE PATCHED AND/OR REPLACED AND FINISHED TO MATCH ADJACENT AREAS AT NO COST TO THE GOVERNMENT.
- E DEMOLITION OF CIVIL, STRUCTURAL, FIRE PROTECTION, PLUMBING, MECHANICAL, ELECTRICAL, AND TELECOMMUNICATION ITEMS INDICATED ON THE ARCHITECTURAL DRAWINGS IS NOT INTENDED TO INCLUDE ALL NECESSARY DEMOLITION WORK FOR THESE ITEMS. REFER TO CIVIL, STRUCTURAL, FIRE PROTECTION, PLUMBING, MECHANICAL, ELECTRICAL, AND TELECOMMUNICATION DRAWINGS FOR ADDITIONAL INFORMATION.
- F DEMOLITION PLANS, INCLUDING EQUIPMENT SHOWN, IS A VISUAL REPRESENTATION OF THE CONDITION OF THE BUILDING AT THE TIME FIELD WORK WAS CONDUCTED.
- G REPAIR ALL DAMAGED CMU AND CONCRETE FOUNDATION SURFACES TO MATCH EXISTING ADJACENT CONSTRUCTION. REFER TO STRUCTURAL DRAWINGS FOR ADDITIONAL INFORMATION.
- H REMOVE AND SALVAGE ALL SIGNAGE AS REQUIRED FOR RENOVATION, UNLESS OTHERWISE NOTED. REINSTALL EXISTING SIGNAGE IN ORIGINAL LOCATION, UNLESS OTHERWISE NOTED. ALL DAMAGED SIGNAGE MUST BE REPLACED IN KIND.
- I REFER TO PHASING PLAN ON G1003. ROOM 120 RENOVATION TO BE COMPLETED DURING PHASE II. ROOM D13 RENOVATION TO BE COMPLETED DURING PHASE IV.
- J CONTRACTOR IS TO PATCH AND REPAIR ALL WALLS.

KEYNOTES

- 1 DEMOLISH EXISTING VCT FLOORING ALL THE WAY DOWN TO EXISTING CONCRETE SUBFLOOR. PREPARE SUBFLOOR TO RECEIVE NEW FLOORING. PROVIDE SELF LEVELING COMPOUND IF NECESSARY TO ACHIEVE A FLAT SURFACE. REFER TO SPECIFICATION SECTION 09 65 00 FOR MORE INFORMATION ABOUT FLOOR PREPARATION. (+/-6,517 SF TOTAL IN BLDG CEP-209)
- 2 REMOVE RESILIENT WALLBASE (+/- 627 SF IN BUILDING CEP-209)
- 3 REMOVE AND DISPOSE OF ALL SYSTEMS FURNITURE COMPLETE
- 4 PROTECT EXISTING SERVER RACK DURING CONSTRUCTION, REFER TO ELECTRICAL DRAWINGS
- 5 REFER TO ELECTRICAL DRAWINGS REGARDING PROTECTION OF CABLES AND EQUIPMENT DURING DEMOLITION. ALL FINISHES IN LAN ROOM ARE TO REMAIN. NO NEW ARCHITECTURAL WORK IN LAN ROOM.
- 6 PATCH AND REPAIR ALL EXISTING TO REMAIN WALLS. REFER TO ELECTRICAL DEMOLITION PLAN FOR MORE INFORMATION REGARDING DEMOLITION THAT WILL EFFECT WALLS. PREPARE WALLS FOR NEW FINISH. REFER TO FINISH SCHEDULE.
- 7 DOOR TO REMAIN, PROTECT DURING DEMOLITION
- 8 DEMOLISH PORTION OF CMU/GYPSUM WALL AS NECESSARY TO ALLOW FOR PLUMBING TIE IN, REFER TO PLUMBING DRAWINGS FOR PLUMBING LINE PLACEMENT

KEY PLAN



DATE	4/29/25	APPR
SYN	REVISION 01	DESCRIPTION

APPROVED	FOR COMMANDER NAVFAC	ACTIVITY	NAVSEA	SATISFACTORY TO DATE	5/6/24
DES	SLG	DRW	SLG	CHK	ALP
PMIDM					SLG
BRANCH MANAGER					ALP
CHIEF ENGINEER	JOHN MILLER, P.E.				
FIRE PROTECTION					

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC FMD NORFOLK
NAVAL STATION NORFOLK
NORFOLK, VIRGINIA

BUILDINGS CEP-200 & CEP-209
FMB OFFICE RENOVATION

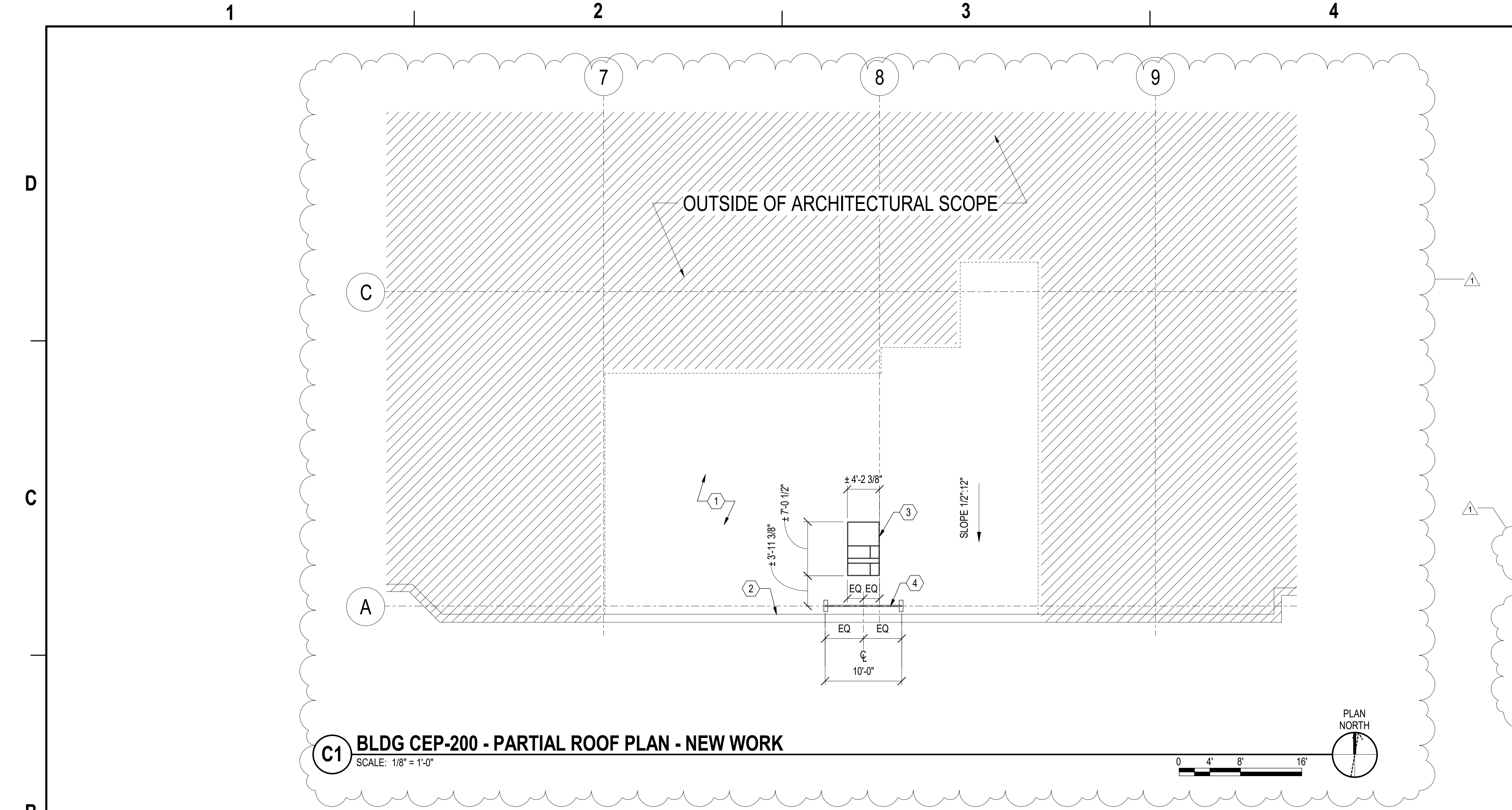
BLDG CEP-209 - PARTIAL FLOOR PLAN - DEMOLITION

SCALE	AS NOTED
PROJECT NO.	1637041
CONSTR. CONTR. NO.	N/A
NAVFAC DRAWING NO.	12.902.247
SHEET	12 OF 78

AD102

DRAWING REVISION: 25 AUGUST 2020

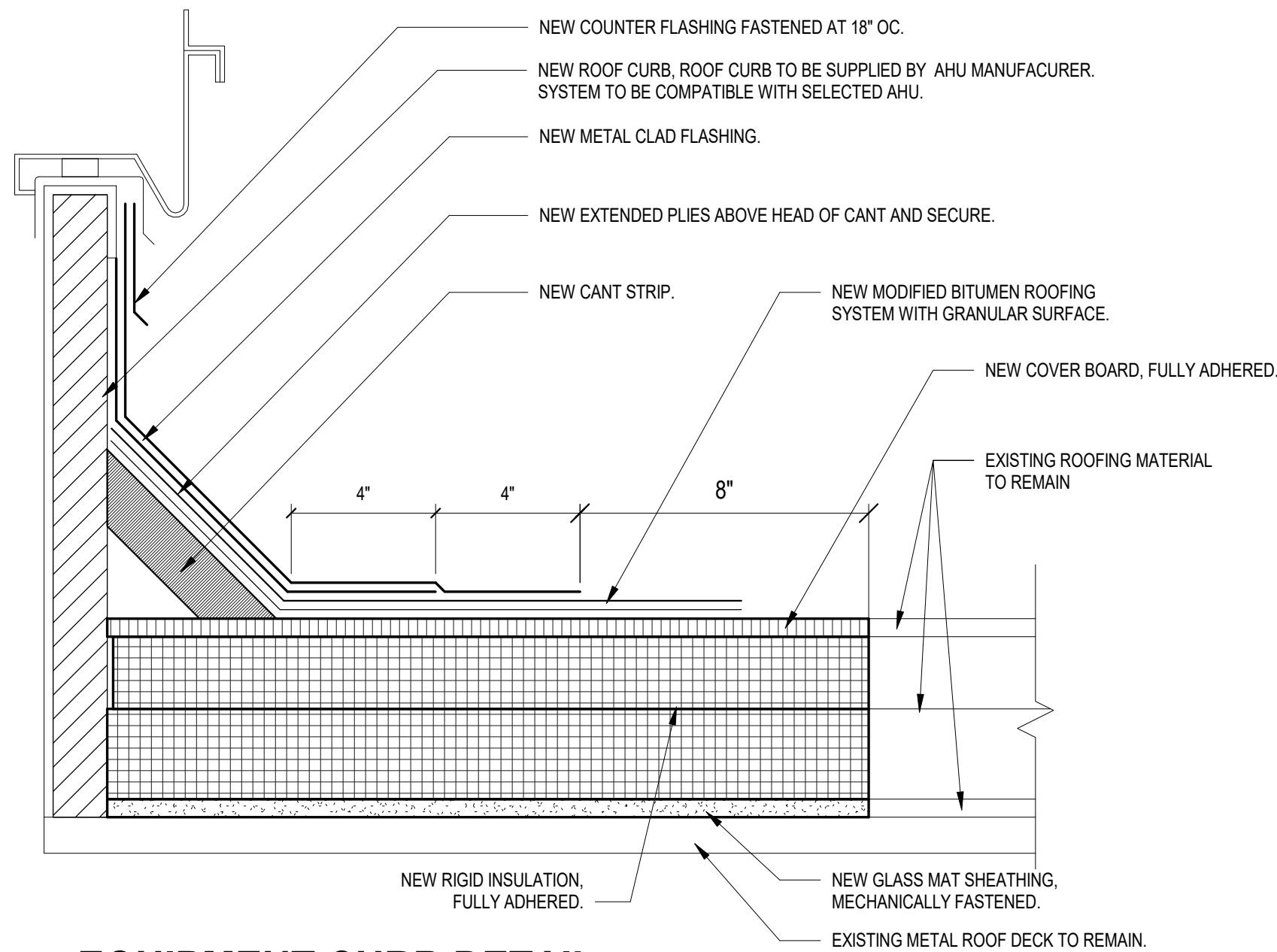
4/29/2025 4:30:51 PM AE103
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C1 BLDG CEP-200 - PARTIAL ROOF PLAN - NEW WORK
SCALE: 1/8" = 1'-0"



A1 CEP200 ROOF IMAGE
SCALE: 12" = 1'-0"



A3 EQUIPMENT CURB DETAIL
SCALE: 3" = 1'-0"

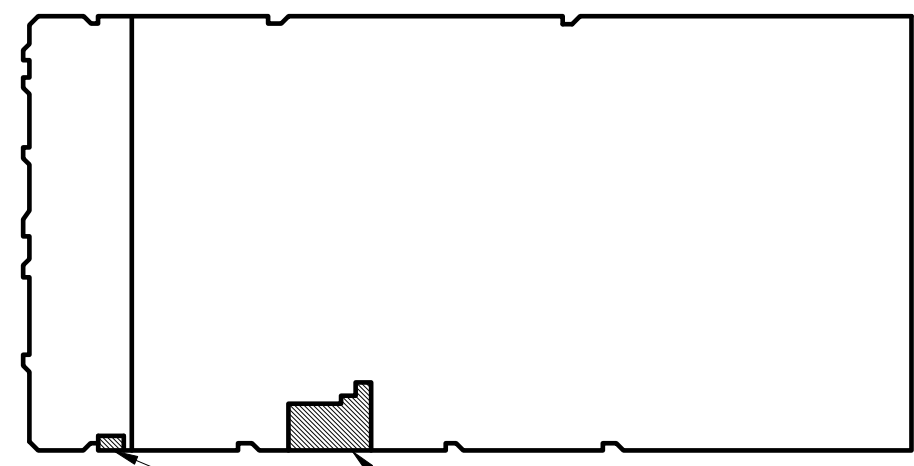
GENERAL SHEET NOTES

- A REFER TO SHEET G1003 FOR PHASING PLAN.
- B REFER TO SHEET A-001 FOR GENERAL ARCHITECTURAL SHEET NOTES.
- C REFER TO MECHANICAL DRAWINGS FOR ADDITIONAL INFORMATION.

KEYNOTES

- 1 EXISTING MODIFIED BITUMEN ROOF - TOP OF ROOF ABOVE PROJECT SPACE IS ± 19'-4" AFF.
- 2 EXISTING PARAPET WALL
- 3 CONTRACTOR TO INSTALL NEW MECHANICAL UNIT ROOF CURB AND NEW MECHANICAL UNIT. NEW OPENING IN ROOF IS REQUIRED. REFER TO MECHANICAL AND STRUCTURAL DRAWINGS. REFER TO EQUIPMENT CURB DETAIL A3/AE103
- 4 CONTRACTOR TO PROVIDE DEMOUNTABLE RAILING SYSTEM THAT IS IN ACCORDANCE WITH EMC 304.11 AND OSHA CFR 1926.502.

KEY PLAN



CEP 200 KEYPLAN

APPR	
4/29/25	DATE
SYM DESCRIPTION	
1	REVISION 01
NAVFAC	
APPROVED	
FOR COMMANDER NAVFAC	
ACTIVITY	
NAVSEA	
SATISFACTORY TO DATE	
DES	SLG
BRW	SLG
CHK	ALP
PM/DW	
SLG	
BRANCH MANAGER	
ALP	
CHIEF ENDORSE	
JOHN MILLER, P.E.	
FIRE PROTECTION	
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND	
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC	
NAVAL STATION NORFOLK	
NORFOLK, VIRGINIA	
BUILDINGS CEP-200 & CEP-209	
FMB OFFICE RENOVATION	
BLDG CEP 200 ROOF PLAN - NEW	
SCALE: AS NOTED	
EPROJCT NO.: 1637041	
CONSTR. CONTR. NO. N/A	
NAVFAC DRAWING NO. 12,902,252	
SHEET	17
OF	78
AE103	
DRAWFORM REVISION: 25 AUGUST 2020	



(A1)

SCALE: 3/32" = 1'-0"

GENERAL SHEET NOTES

- A REFER TO SHEET G1003 FOR PHASING PLAN.
- B REFER TO SHEET A-001 FOR GENERAL ARCHITECTURAL SHEET NOTES.
- C REFER TO MECHANICAL DRAWINGS FOR ADDITIONAL INFORMATION.

(THIS SHEET ONLY)

- 1 UPPER ROOF OUTSIDE OF SCOPE OF WORK
- 2 MINI SPLIT CONDENSER UNIT - REFER TO MECHANICAL DRAWINGS FOR ROOF MOUNTING DETAILS
- 3 EXISTING MODIFIED BITUMEN ROOF - TOP OF ROOF IS $\pm 24'-0"$ AFF
- 4 EXISTING ROOF HATCH



VIEW FROM UPPER ROOF TO LOWER ROOF - TOWARDS
EXISTING ROOF HATCH



VIEW FROM UPPER ROOF - INTERSECTION OF LOWER AND UPPER ROOF

(A4)

SCALE: 1" = 1'-0"

[illegible]

APPROVED		A/E INFO	
			
FOR COMMANDER NAVFAC		5-9-25	
ACTIVITY			
NAVFSEA			
SATISFACTORY TO DATE		5/6/24	
DES	SLG	DRW	SLG
CHK	ALP		
PMDM		SL	
BRANCH MANAGER		AL	
CHIEF ENGINEER		JOHN MILLER, P.	
C/E'S CERTIFICATION			

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
MIDLAND PWD NORFOLK
NAVAL STATION NORFOLK
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL STATION - NORFOLK, VA
NORFOLK VIRGINIA

OLK
BUILDINGS CEP-200 & CEP-209

NGS CEF-200 & CEF-100 FMB OFFICE RENOVATION

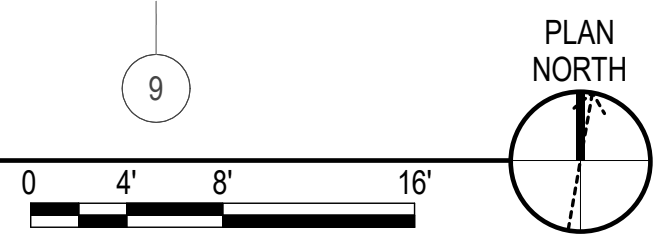
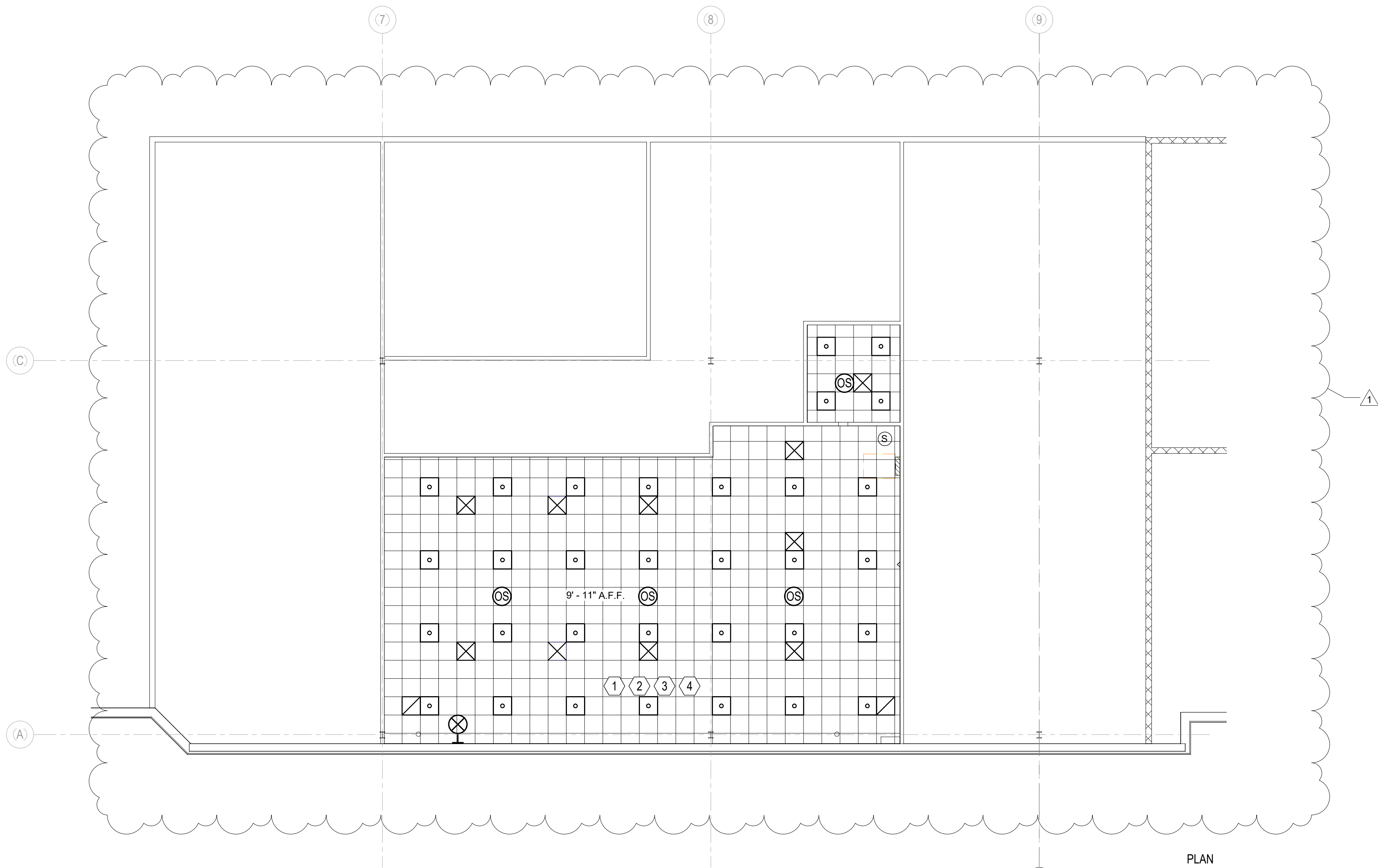
BLDG CEP 209 ROOF PLAN - NEW

SCALE: AS NOTED		
EPROJECT NO.:		1637041
CONSTR. CONTR. NO.		N/A
NAVFAC DRAWING NO.		12,902.253
SHEET	18	OF 78

AE104

4/29/2025 4:32:52 PM AE111
C:\Rent\2021 Local Models\CEP200.209_1637041_A_121_sarah.gip\plan1.rvt

B1 BLDG CEP200 - PARTIAL REFLECTED CEILING PLAN - NEW
SCALE: 1/8" = 1'-0"



GENERAL SHEET NOTES

- A UNLESS OTHERWISE NOTED, CEILING TILE MUST BE CENTERED IN EACH ROOM, IN ACCORDANCE WITH THE LAYOUTS SHOWN.
- B MAKE ADJUSTMENTS NECESSARY TO ELIMINATE SMALL OR ODD PIECES OF CEILING TILE
- C REFER TO FIRE PROTECTION, MECHANICAL AND ELECTRICAL DRAWINGS FOR LIGHT FIXTURES, EXIT LIGHT LOCATIONS, AIR DIFFUSERS, EXHAUST FANS, ETC.
- D WHERE TILES ARE CUT, COPE AND FINISH EDGES TO MATCH UNCUT TILE.
- E CENTER ALL LIGHTS, DIFFUSERS, FIXTURES, SPRINKLER HEADS ETC IN CEILING TILES UNLESS OTHERWISE NOTED.
- F DAMP CLEAN AND HAND DRY ALL FIXTURES AND LAMPS JUST PRIOR TO FINAL INSPECTION.

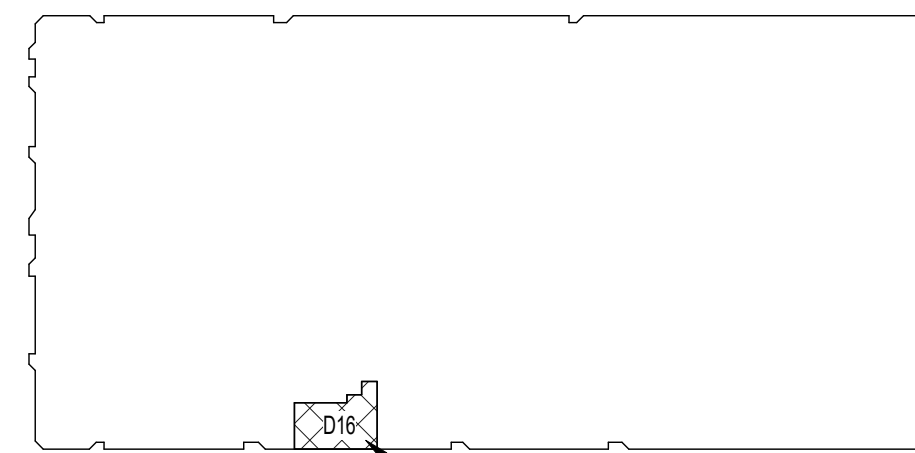
(THIS SHEET ONLY) KEYNOTES

- 1 NEW ACOUSTIC CEILING TILE AND SUPPORTING GRID SYSTEM ($\pm$ 1,955 SF IN BLDG CEP 200)
- 2 REFER TO MECHANICAL, ELECTRICAL, AND FIRE PROTECTION DRAWINGS FOR ADDITIONAL INFORMATION
- 3 NEW LIGHTING AND AIR TERMINAL DEVICES, REFER TO ELECTRICAL AND MECHANICAL DRAWINGS
- 4 PROVIDE EXTRUDED ALUMINUM TRIM PIECE AT PERIMETER OF CEILING GRID - AT INTERSECTION OF WALL AND CEILING ($\pm$ 225 LF ALUMINUM TRIM ANGLE 1" X 1 1/2")

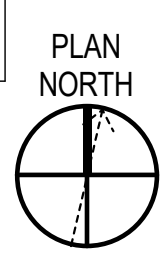
NEW REFLECTED CEILING PLAN LEGEND

SYMBOL	DESCRIPTION
	AREA NOT IN CONTRACT
	2' - 0" X 2' - 0" NEW ACOUSTICAL CEILING TILE
X'-X" A.F.F.	FINISHED CEILING HEIGHT ABOVE FINISHED FLOOR
	2' - 0" X 2' - 0" NEW LIGHT FIXTURE - REFER TO ELECTRICAL DRAWINGS
OR	NEW EXIT SIGNS. REFER TO ELECTRICAL DRAWINGS.
	NEW OCCUPANCY SENSOR. REFER TO ELECTRICAL DRAWINGS.
	NEW SPEAKER. REFER TO ELECTRICAL DRAWINGS.
	NEW SUPPLY DIFFUSER. REFER TO MECHANICAL DRAWINGS.
	NEW CEILING RETURN OR EXHAUST. REFER TO MECHANICAL DRAWINGS.

KEY PLAN



BUILDING CEP-200 AREA OF WORK



APPROVED	DATE	4/29/25	APPR
FOR COMMANDER NAVFAC	DATE	4/29/25	APPR
ACTIVITY	SYMBOL	DESCRIPTION	DATE
NAVSEA	1	REVISION 01	4/29/25
SATISFACTORY TO DATE	5/6/24		
DES	SLG	BRW	SLG
CHK	ALP		
PMIDM	SLG		
BRANCH MANAGER	ALP		
CHIEF ENGINEER	JOHN MILLER, P.E.		
FIRE PROTECTION			
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND			
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC			
MID-ATLANTIC			
NAVAL STATION - NORFOLK, VA			
NAVAL STATION NORFOLK			
NORFOLK, VIRGINIA			
BUILDINGS CEP-200 & CEP-209			
FMB OFFICE RENOVATION			
BLDG CEP 200 - PARTIAL FLOOR PLAN REFLECTED CEILING PLAN - NEW			
SCALE: AS NOTED			
PROJECT NO.:	1637041		
CONSTR. CONTR. NO.	N/A		
NAVFAC DRAWING NO.	12,902,254		
SHEET	19	OF	78
AE111			
DRAWING REVISION: 25 AUGUST 2020			

GENERAL ELECTRICAL NOTES

1. ALL ELECTRICAL WORK SHALL BE ACCOMPLISHED IN ACCORDANCE WITH THE NFPA 70 NATIONAL ELECTRICAL CODE 2023.
2. CONTRACTOR MUST COORDINATE CONNECTIONS TO POWER, DATA, AND TELEPHONE DISTRIBUTION SYSTEMS WITH CONTRACTING OFFICE PRIOR TO BEGINNING WORK.
3. PROVIDE 208Y/120V FOR DISTRIBUTION POWER TO TRAILER PANELS. PROVIDE MAIN DISTRIBUTION PANEL SIZED 800A FOR BOTH TRAILERS IN MAIN ELECTRICAL ROOM OF TRAILER A AND CONNECT TO 225KVA PAD MOUNTED TRANSFORMER. PROVIDE MAIN DISTRIBUTION PANEL IN TRAILER B, SIZED 400A AND CONNECT TO MAIN PANEL IN TRAILER A. ALL DISTRIBUTION PANELS MUST BE FED FROM MAIN DISTRIBUTION PANELS. MAIN DISTRIBUTION PANELS IN TRAILERS A AND B MUST HAVE AN APPROPRIATELY SIZED DISCONNECT SWITCH OR MAIN CIRCUIT BREAKER ON INCOMING FEEDER LOCATED WITHIN THE SAME ROOM AS THE PANEL.
4. PROVIDE LED RECESSED TROFFER LIGHT FIXTURES IN OPEN OFFICE AREA. PROVIDE MINIMUM OF 30 FC AT 2'-6" AFF. PROVIDE MANUAL LIGHTING CONTROLS AND OCCUPANCY SENSORS. PROVIDE SURFACE MOUNTED LED LINEAR WRAP FIXTURES IN ALL ELECTRICAL AND TELECOM ROOMS TO PROVIDE A MINIMUM OF 50 FC AT 2'-6" AFF. PROVIDE MANUAL LIGHTING CONTROLS ONLY IN THESE SPACES. PROVIDE 3 LED DOWNLIGHT FIXTURES CENTERED ABOVE RESTROOM SINKS WITH LIGHT CONTROL SWITCH LOCATED NEXT TO SINK.
5. PROVIDE GENERAL PURPOSE RECEPTACLES MOUNTED AROUND THE TRAILER INTERIOR WALLS SO NO POINT ALONG THE WALL IS MORE THAN 12' APART. LOCATE RECEPTACLES TO AVOID BEING COVERED BY FURNITURE SYSTEMS. PROVIDE QUAD RECEPTACLES ON EACH INTERIOR WALL OF TELECOMMUNICATIONS AND ELECTRICAL ROOMS. PROVIDE
6. PROVIDE POWER FOR SYSTEMS FURNITURE VIA FURNITURE WHIP OR POWER POLE. SYSTEMS FURNITURE IS A 4-2-2 CIRCUIT CONFIGURATION, 4 HOT WIRES, 2 NEUTRAL, AND 2 GROUND, RUN FROM 2-2 POLE CIRCUIT BREAKERS WITH HANDLE TIE. NO MORE THAN 4 WORKSTATIONS CAN BE FED FROM 1 CIRCUIT. PROVIDE GENERAL PURPOSE RECEPTACLES ON OPEN WALLS WHERE NOT CONCEALED BY SYSTEMS FURNITURE, SPACED NO MORE THAN 15' FROM THE NEXT NEAREST RECEPTACLE OR SYSTEMS FURNITURE, MINIMUM OF 3 RECEPTACLES PER TRAILER.
7. PROVIDE POWER TO MECHANICAL UNITS AS PER MANUFACTURER'S REQUIREMENTS. COORDINATE WITH MECHANICAL PRIOR TO INSTALLATION.
8. PROVIDE 2-JACK DATA OUTLET AT EACH WORKSTATION. PROVIDE CAT6 CABLE FROM OUTLET TO TELECOM ROOM VIA FURNITURE WHIP OR POWER POLE. PROVIDE ADDITIONAL 6' SLACK OF CABLE IN TELECOMMUNICATIONS ROOM, COIL, AND LABEL FOR CONNECTION BY OTHER.
9. PROVIDE 1-JACK TELEPHONE OUTLET AT EACH WORKSTATION. PROVIDE CAT6 CABLE FROM OUTLET TO TELECOM ROOM VIA FURNITURE WHIP OR POWER POLE. PROVIDE ADDITIONAL 6' SLACK OF CABLE IN TELECOMMUNICATIONS ROOM, COIL, AND LABEL FOR CONNECTION BY OTHER.
9. PROVIDE POWER FOR RESTROOM HAND DRYERS AS PER MANUFACTURER'S REQUIREMENTS. COORDINATE PRIOR TO INSTALLATION.
10. PROVIDE EWC RECEPTACLE FOR ELECTRIC WATER COOLER. COORDINATE EXACT LOCATION PRIOR TO INSTALLATION.

GENERAL STRUCTURAL NOTES

THE CONTRACTOR SHALL RETAIN THE SERVICES OF A PROFESSIONAL STRUCTURAL/GEOTECHNICAL ENGINEER, WHO WILL CONFER WITH THE SELECTED TRAILER MANUFACTURER TO DESIGN A FOUNDATION AND TIE-DOWN SYSTEM FOR EACH TRAILER. THE SELECTED ENGINEER SHALL POSSESS A VALID LICENSE TO PRACTICE IN THE COMMONWEALTH OF VIRGINIA, AND SHALL HAVE A MINIMUM OF 5 YEARS' EXPERIENCE DESIGNING FOUNDATIONS IN COASTAL VIRGINIA. THE SELECTED ENGINEER SHALL PROVIDE A STAMPED/SEALED FOUNDATION DESIGN CONSISTING OF LOAD CALCULATIONS AND PERTINENT DRAWINGS AND/OR SKETCHES TO THE CONTRACTING OFFICER FOR REVIEW AND APPROVAL. GOVERNMENT APPROVAL OF THE CONTRACTOR'S FOUNDATION DESIGN IS REQUIRED PRIOR TO INSTALLING THE PROPOSED FOUNDATION SYSTEM.

2. ALL TRAILERS/FOUNDATIONS WILL BE DESIGNED ASSUMING AN OCCUPANCY/RISK CATEGORY OF II.

3. TRAILER FOUNDATIONS SHALL BE DESIGNED USING THE FOLLOWING PARAMETERS:

- a. BEARING CAPACITY OF EXISTING SOILS MAY NOT EXCEED AN ALLOWABLE BEARING PRESSURE OF 1,500 PSF.
- b. FACTOR OF SAFETY AGAINST SLIDING: 2.0.
- c. FACTOR OF SAFETY AGAINST OVERTURNING: 2.0.

4. LOADS TO BE USED FOR BEARING CAPACITY DESIGN OF ALL FOUNDATIONS ARE AS FOLLOWS:

DEAD LOAD - FURNISHED TRAILER (UON BY TRAILER MFR) -----50 PSF

FLOOR LIVE LOAD (OFFICE/CLASSROOM DESIGNATION) -----50 PSF

ROOF LIVE LOAD -----20 PSF

SNOW DESIGN CRITERIA: GROUND SNOW LOAD (PG) -----
----- (ASCE 7-16) 10 PSF
----- (ASCE 22) 36 PSF

SNOW EXPOSURE FACTOR (CE) -----1.0

SNOW LOAD IMPORTANCE FACTOR (IS) -----1.0

5. LOADS TO BE USED FOR SLIDING, UPLIFT AND OVERTURNING CAPACITY OF ALL FOUNDATIONS ARE AS FOLLOWS:

DEAD LOAD - UNFURNISHED TRAILER (UON BY TRAILER MFR) -----30 PSF

WIND DESIGN CRITERIA (PER ASCE 7-16 OR 22):

BASIC WIND SPEED (VULT) -----119 MPH

BASIC WIND SPEED (VASD) -----92.2 MPH

WIND LOAD IMPORTANCE FACTOR (IW) -----1.0

WIND EXPOSURE CATEGORY -----C

ENCLOSURE CLASSIFICATION -----N/A FOR FOUNDATION DESIGN

IN ADDITIONAL TO THE APPLICATION OF HORIZONTAL WIND PRESSURES ACTING ON THE SIDE OF ANY GIVEN TRAILER, WIND UPLIFT PRESSURES MUST ALSO BE APPLIED TO THE ENTIRE BOTTOM SURFACE OF EACH TRAILER USING A RATIONAL METHODOLOGY, IN THE SAME FASHION THAT UPLIFT WOULD BE APPLIED TO EITHER A STAND-ALONE OPEN CANOPY STRUCTURE, OR TO A FREESTANDING TANK OR PIECE OF EQUIPMENT.

6. IN ORDER TO MINIMIZE THE AMOUNT OF GROUND DISTURBANCE/EXCAVATION ASSOCIATED WITH THIS PROJECT, THE USE OF EXCAVATED FOOTINGS OR CAST-IN-PLACE CONCRETE FOOTINGS IS PROHIBITED. ALL FOUNDATIONS MUST CONSIST OF EITHER PRECAST CONCRETE ELEMENTS, MASONRY PIERS, SLAB ON GRADE OR LIFTED CONCRETE STEM WALL FOUNDATION, HOT-DIP GALVANIZED STEEL CRIBBING, OR MFR'S PROPRIETARY FOUNDATION SYSTEMS INSTALLED DIRECTLY UPON EXISTING GRADE.

7. MANUFACTURER'S LITERATURE REGARDING THE PULLOUT CAPACITY/STRENGTH OF ANY TIE-DOWN ANCHORS PROPOSED FOR USE ON THIS PROJECT MUST BE SUPPORTED BY THE RESULTS OF MANUFACTURER FIELD TESTING WHICH INDICATES PROVEN STRENGTH CAPABILITIES UNDER SIMILAR LOADING CONDITIONS, AND IN SIMILAR SOIL CONDITIONS, AS THOSE PRESENT HEREIN; OR SHALL OTHERWISE BE BASED UPON THE SUCCESSFUL TESTING OF SIMILAR/IDENTICAL SOIL ANCHORAGE SYSTEMS WHERE SUCH TEST RESULTS HAVE BEEN PUBLISHED IN A REPUTABLE COMMERCIAL ENGINEERING JOURNAL, OR GOVERNMENT OR UNIVERSITY PUBLICATION. UNTESTED, UNPROVEN SYSTEMS WILL NOT BE ACCEPTED.

GENERAL MECHANICAL NOTES

1. ALL PLUMBING WORK SHALL BE ACCOMPLISHED IN ACCORDANCE WITH THE INTERNATIONAL PLUMBING CODE 2009 EDITION.
2. ALL MECHANICAL WORK SHALL BE ACCOMPLISHED IN ACCORDANCE WITH THE INTERNATIONAL MECHANICAL CODE 2009 EDITION AND THE LATEST ASHRAE 62.1.
3. PROVIDE A COMPLETE AND OPERABLE AIR CONDITIONING/ HEATING SYSTEM WITH SUPPLY/RETURN DUCTWORK, CEILING DEVICES, AND THERMOSTATS. DO NOT RETURN AIR FROM ROOMS BEING EXHAUSTED. PROVIDE EXHAUST FOR KITCHENS RESTROOMS AND JANITORS CLOSET.
4. PROVIDE DEDICATED A/C UNIT FOR NMCI CLOSET AND SERVER ROOMS.
5. DOMESTIC COLD WATER AND HOT WATER PIPING SHALL BE EITHER CPVC PLASTIC PIPE OR COPPER TYPE "L" PROVIDE WITH 1" FIBERGLASS INSULATION WITH ALL SERVICE JACKET. SUPPORT PIPING FROM STRUCTURE WITH MANUFACTURED TYPE HANGERS AND INSULATION SHIELDS.
6. BUILDING WASTE, VENT AND DRAIN PIPING SHALL BE DWV SCHEDULE 40 PLASTIC PIPING. SUPPORT PIPING FROM STRUCTURE WITH MANUFACTURED TYPE HANGERS.
7. PROVIDE A/C UNITS WITH DUCTED RETURNS IN SERVER ROOMS.
8. BUILDING WASTE MAINS MUST BE A MINIMUM OF 4" AND BE PROVIDED WITH LONG RADIUS ELBOWS. URINALS AND WATER CLOSETS MUST BE COMMERCIAL GRADE FLUSH VALVE TYPE. FLUSH TANKS ARE NOT PERMITTED.
9. PROVIDE PLACARD ABOVE EACH THERMOSTAT LABELLING ITS CORRESPONDING HVAC UNIT.

GENERAL SIGNAGE NOTES

1. THE CONTRACTOR SHALL VERIFY ALL CONDITIONS AND DIMENSIONS BEFORE PROCEEDING.
2. THE CONTRACTOR IS RESPONSIBLE FOR SITE VERIFICATION PRIOR TO ORDERING AND INSTALLATION OF ALL ITEMS DESCRIBED ON THE PLANS AND IN THE SPECIFICATIONS AND ANY ADDITIONAL HARDWARE OR SIMILAR ITEMS TO COMPLETE INSTALLATION OF SIGNAGE.
3. THE WORK SHALL CONFORM TO ALL APPLICABLE CODES AND REGULATIONS.
4. WORK SHALL INCLUDE ALL LABOR AND MATERIALS NECESSARY FOR A COMPLETE AND FINISHED INSTALLATION READY FOR THE INTENDED USE.
5. TYPICAL SIGNAGE IS BASED ON "ESSENTIALS" SIGNAGE FROM 2/90 SIGN SYSTEMS. FINISHES ARE LISTED ON IN601.
6. ALL SIGN TYPES SHALL MEET AMERICANS WITH DISABILITIES ACT (ADA) REQUIREMENTS.
7. SIGN MOUNTING INFORMATION IS SHOWN ON AG601.
8. USE D-SCREW MOUNTING METHOD ON SIGNS MOUNTED ON WALLS. USE PRESSURE SENSITIVE TAPE ON SIGNS MOUNTED ON DOORS.

GENERAL SHEET NOTES

1. CONTRACTOR TO MAKE ALL UTILITY CONNECTIONS INCLUDING: WATER, SANITARY, ELECTRICAL AND TELEPHONE- COORDINATE WITH SITE PLAN.
2. CONTRACTOR TO INCLUDE TRAILER ANCHORING, SKIRTING, RAISED WALKWAY BETWEEN TRAILERS, STEP ASSEMBLIES AND IBC COMPLIANT RAMP SYSTEM.
3. NOT USED.
4. THE PLANS SHOWN ON THIS SHEET ARE FOR DIAGRAMMATIC ILLUSTRATION REFLECTING THE PROGRAM FOR DESIGN. SHOWING THE FURNITURE AND FLOOR PLAN LAYOUT ONLY. MODULAR BUILDING MANUFACTURER TO SUPPLY FURNITURE.
5. THE FINAL DESIGN OF THE TEMPORARY MODULAR BUILDINGS SHALL BE THE RESPONSIBILITY OF THE MODULAR BUILDING MANUFACTURER.
6. THE PLAN FOR THE TEMPORARY MODULAR BUILDINGS ARE BASED ON TRAILERS 11'-9" X 36'-0" AND 11'-9" X 48'-0".
7. FINAL LAYOUT FROM THE CONTRACTOR SHALL MAINTAIN THE AREAS AND ROOM PROPORTIONS SHOWN.
8. REFER TO CIVIL DRAWINGS FOR TEMPORARY MODULAR BUILDING LOCATION, ACCESS, AND SITE UTILITY DETAILS.
9. FOUNDATION DESIGN SHALL BE THE RESPONSIBILITY OF THE MODULAR BUILDING MANUFACTURER AND COORDINATED WITH THE GENERAL CONTRACTOR.
10. REFER TO SPECIFICATION SECTION 13 12 00 FOR FURTHER INFORMATION.

NOTE: ACTUAL MANUFACTURED TRAILER MAY HAVE SLIGHTLY DIFFERENT OVERALL AND INTERIOR DIMENSIONS.

OCCUPANCY/FIXTURE COUNT

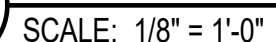
LIFE SAFETY OCCUPANCY

TRAILERS ARE A "BUSINESS OCCUPANCY" THEREFORE 150
GSF/PERSON.
TOTAL = 6,345 GSF/150 GSF = 43

PLUMBING FIXTURE COUNT
43 OCCUPANTS
FIXTURES PER CODE - 1 PER 25 FOR THE FIRST 50 AND 1
PER 50 FOR THE REMAINDER EXCEEDING 50

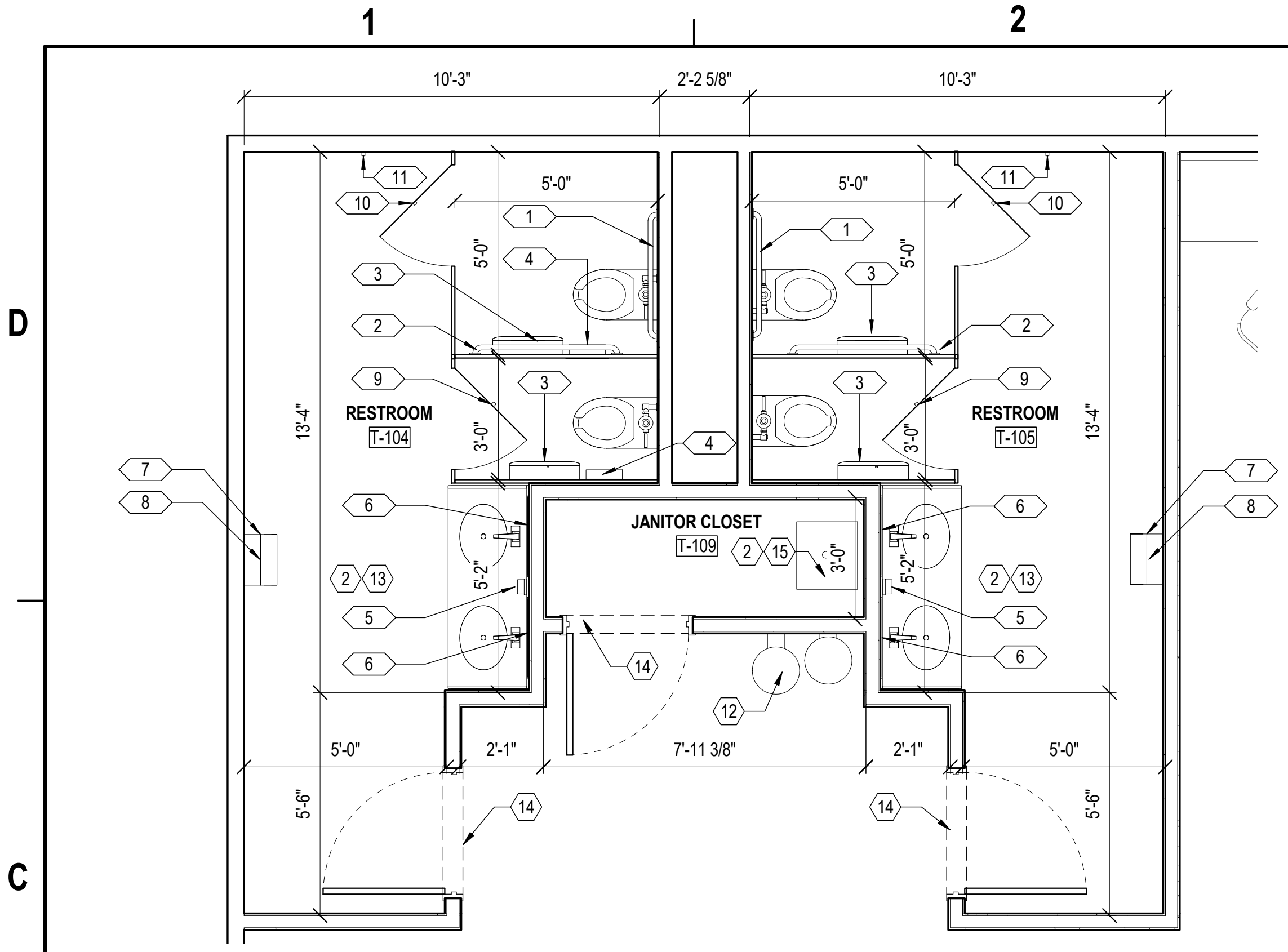
NUMBER OF REQUIRED TOILET FIXTURES
MALE: 1 FIXTURES; 1 LAVATORIES
FEMALE: 1 FIXTURES; 1 LAVATORIES
1 DRINKING FOUNTAINS
1 MOP SINK

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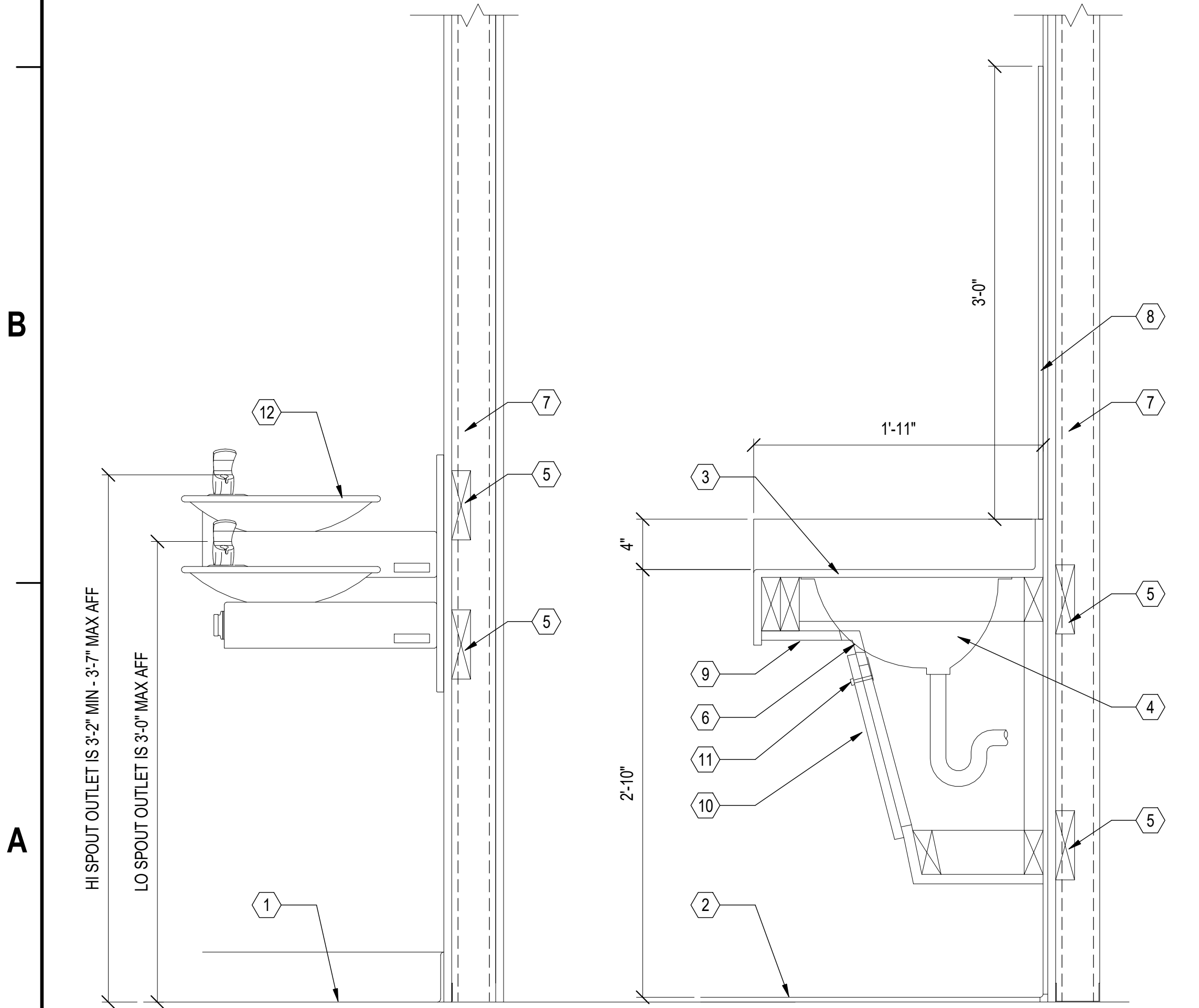


DRAWFORM REVISION: 25 AUGUST 2020

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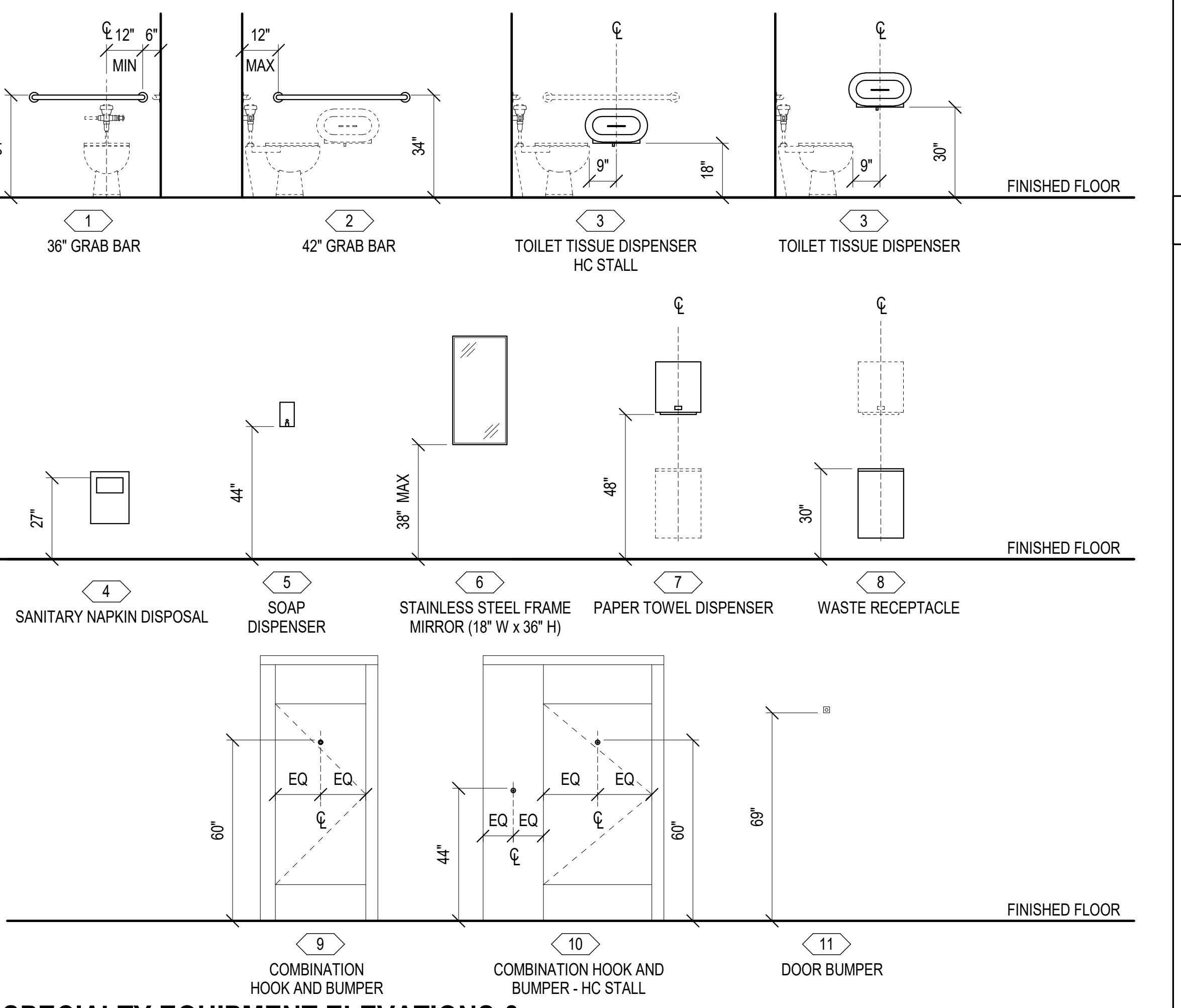
C1 TRAILER B - ENLARGED BATHROOM PLAN
SCALE: 3/8" = 1'-0" (AE122)



A1 TRAILER DETAILS
SCALE: 1 1/2" = 1'-0"

SPECIALTY EQUIPMENT SCHEDULE				
MARK	DESCRIPTION	MANUFACTURER	MODEL NO.	REMARKS
1	36" GRAB BAR	BOBRICK	B-6806 SERIES	1, 4, 5, 6, 7, 8
2	42" GRAB BAR	BOBRICK	B-6806 SERIES	1, 4, 5, 6, 7, 8
3	SURFACE-MOUNTED TWIN JUMBO- ROLL TOILET PAPER DISPENSER	BOBRICK	B-2892	1, 5, 8
4	NAPKIN DISPOSAL	BOBRICK	B-353	1, 4, 8
5	SOAP DISPENSER	BOBRICK	B-2111	3, 8
6	24 X 36 STAINLESS STEEL FRAME MIRROR	BOBRICK	B-165-2436	2, 4, 8
7	PAPER TOWEL DISPENSER (GFCI)	KIMBERLY-CLARK	KCC-09990	1, 4, 8
8	WASTE RECEPTACLE	BOBRICK	B-277 CONTURA SERIES	1, 4, 8
9	COMBINATION HOOK AND BUMPER	BOBRICK	B-9541	1,4,8
10	COMBINATION HOOK AND BUMPER	BOBRICK	B-9541	1,4,8
11	COMBINATION HOOK AND BUMPER	BOBRICK	B-9541	1,4,8
T	290 FIRE EXTINGUISHER SAFETY SIGN			

SPECIALTY EQUIPMENT SCHEDULE REMARKS	
1	SATIN FINISH
2	CENTER MIRROR OVER THE SINK, ALIGN BOTTOM OF MIRROR WITH TOP OF BACKSPLASH
3	CENTER BETWEEN MIRRORS
4	REFER TO AE122 FOR MOUNTING TYPICAL MOUNTING HEIGHTS
5	PROVIDE WOOD BLOCKING OR 18 GA X 6" WIDE METAL PLATE IN WALL AS REQUIRED TO SECURE SPECIALTY EQUIPMENT
6	GRAB BAR MUST HAVE PEENED (NON-SLIP) GRIPPING SURFACE
7	MOUNT WITH LONG DIMENSION VERTICAL
8	SPECIALTY EQUIPMENT TO BE PROVIDED BY THE MODULAR TRAILER MFR.



A2 SPECIALTY EQUIPMENT ELEVATIONS & MOUNTING HEIGHTS
SCALE: 3/8" = 1'-0"

- ### GENERAL SHEET NOTES
- A CONTRACTOR TO MAKE ALL UTILITY CONNECTIONS INCLUDING WATER, SANITARY, ELECTRICAL AND TELEPHONE - COORDINATE WITH SITE PLAN.
 - B CONTRACTOR TO INCLUDE TRAILER ANCHORING, SKIRTING, RAISED WALKWAY BETWEEN TRAILERS, STEP ASSEMBLIES AND IBC COMPLIANT RAMP SYSTEM.
 - C NOT USED.
 - D THE PLANS SHOWN ON THIS SHEET ARE FOR DIAGRAMATIC ILLUSTRATION REFLECTING THE PROGRAM FOR DESIGN, SHOWING THE FURNITURE AND FLOOR PLAN LAYOUT ONLY.
 - E THE FINAL DESIGN OF THE TEMPORARY MODULAR BUILDINGS SHALL BE THE RESPONSIBILITY OF THE MODULAR BUILDING MANUFACTURER.
 - F THE PLAN FOR THE TEMPORARY MODULAR BUILDINGS ARE BASED ON TRAILERS 11'-9" X 36'-0".
 - G THE FINAL LAYOUT FROM THE CONTRACTOR SHALL MAINTAIN THE AREAS AND ROOM PROPORTIONS SHOWN.
 - H REFER TO CIVIL DRAWINGS FOR TEMPORARY MODULAR BUILDING LOCATION, ACCESS AND SITE UTILITY DETAILS.
 - I FOUNDATION DESIGN SHALL BE THE RESPONSIBILITY OF THE MODULAR BUILDING MANUFACTURER AND COORDINATED WITH THE GENERAL CONTRACTOR.
 - J REFER TO SPECIFICATION SECTION 13 12 00 FOR FURTHER INFORMATION.
 - K MANUFACTURER'S TRADE NAMES AND NUMBERS LISTED ARE NOT INTENDED TO BE PROPRIETARY. PRODUCTS OF OTHER MANUFACTURERS WILL BE ACCEPTABLE, PROVIDED THEY MATCH THE COLORS, FINISHES, TEXTURES AND PATTERNS OF THE MANUFACTURERS LISTED, AND MEET THE TECHNICAL REQUIREMENTS OF THE SPECIFICATIONS.

- ### KEYNOTES
- (THIS SHEET ONLY)
- 1 TRAILER SUB-FLOOR
 - 2 CONTINUOUS RESILIENT FINISH FLOOR
 - 3 SOLID SURFACE COUNTERTOP WITH INTEGRAL BACKSPLASH
 - 4 UNDER COUNTER MOUNTED SINK, REFER TO PLUMBING
 - 5 CONTINUOUS TREATED WOOD BLOCKING
 - 6 NOTCH PANEL FOR SINK PLACEMENT
 - 7 WALL AS SCHEDULED
 - 8 MIRROR, REFER TO SPECIALTY EQUIPMENT SCHEDULE
 - 9 PLASTIC LAMINATE ON ALL EXPOSED SURFACES
 - 10 PLASTIC LAMINATE DOOR
 - 11 SLOTTED QUARTER-TURN LATCH, CLEAR ALUMINUM FINISH
 - 12 HI-LOW ACCESSIBLE WATER FOUNTAIN
 - 13 FRP PANEL, MARLITE SYMMETRIX. SELECT FROM MANUFACTURER STANDARD COLOR SELECTION. FRP ON ALL WALL, RUN FRP FULL HEIGHT FROM FINISH FLOOR TO CEILING
 - 14 THRESHOLD AT RESTROOMS
 - 15 MOP SINK

APPROVED	DATE	4/29/25	APPR
FOR COMMANDER NAVFAC	DESCRIPTION	1	REVISION 01
ACTIVITY	SYN	1	REVISION 01
NAVSEA	DATE	5/6/24	ALP
DES	SLG	DRW	SLG
PMID	SLG	CHK	ALP
BRANCH MANAGER	ALP		
CHIEF ENGINEER	JOHN MILLER, P.E.		
FIRE PROTECTION			
DEPARTMENT OF THE NAVY	NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND		
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC	NAVAL STATION - NORFOLK, VA		
NAVAL STATION NORFOLK	NORFOLK, VIRGINIA		
BUILDINGS CEP-200 & CEP-209	FMB OFFICE RENOVATION		
ENLARGED TRAILER PLANS - NEW			
SCALE: AS NOTED	PROJECT NO. 1637041		
EPROJCT NO. 1637041	CONSTR. CONTR. NO. N/A		
NAVFAC DRAWING NO. 12,902,258	SHEET 23 OF 78		
AE123			
DRAWING REVISION: 25 AUGUST 2020			

DESIGN CONDITIONS

	SUMMER		WINTER
	°FDB	°FWB	°FDB
OUTSIDE	91	77	22
INSIDE	68	62.4	68

GRILLE, REGISTER, AND DIFFUSER SCHEDULE

MARK	SERVICE	CFM RANGE	MAX. NC	MAX. SP (IN. WG)	NECK SIZE	SHELL SIZE	TYPE	THROW FT. (AT 50 FPM TERMINAL VEL.)	REMARKS
A	SUPPLY	185 - 275	25	.10	8"Ø	24 x 24	1	15	1,3,5,7
B	SUPPLY	280 - 490	25	.10	10"Ø	24 x 24	1	17	1,3,5,7
C	SUPPLY	500 - 680	25	.10	12"Ø	24 x 24	1	20	1,3,5,7
Y	RETURN	1050	25	.05	20 x 20	24 x 24	9	-	2,3,6,7
Z	RETURN	1050	25	.05	20 x 20	24 x 24	2	-	2,3,6,7,8

DUCT CONSTRUCTION AND LEAK TEST SCHEDULE

[illegible]

AIR HANDLING UNIT SCHEDULE

MARK	LOCATION	SUPPLY FAN DATA				HEATING				DX COOLING COIL DATA						FILTER TYPE	ELECTRICAL DATA					REFRIGERANT	UNIT MIN SEER	REMARKS	
		SA CFM	OA CFM	ESP (IN WG.)	HP	CFM	EAT ("FDB)	HEAT PUMP		ELEC. COIL	TOTAL (MBH)	SENSIBLE (MBH)	EAT		LAT										
								CAP.(MBH)	LAT ("FDB)	KW			"FDB	"FWB	"FDB		"FWB								
AHU-1	ROOF	2400	300	0.7	1.5	2400	60	33.7	81.3	15.0	65.5	51.4	80.0	67.0	59.2	58.2	N/A	32	45	208	3	60	R-410A	12.1	2,3,4,5

DUCTLESS SPLIT-SYSTEM CONDENSING UNIT SCHEDULE

TAG	TYPE	ELECTRICAL DATA				FLA	AIR ENT. CONDENSER ("FDB)	NOTES
		VOLTS	PHASE	MCA	MFS			
CU-1	COOLING ONLY	208	1	19.0	26	.75	95	1,2,3,4,5,6
CU-2	COOLING ONLY	208	1	19.0	26	.75	95	1,2,3,4,5,6

SEQUENCE OF OPERATION

AHU-1

COOLING:

WHEN THE SYSTEM IS INDEXED TO COOLING AND THE SPACE TEMPERATURE RISES ABOVE THE SPACE TEMPERATURE SETPOINT, INITIALLY SET AT 68°F (ADJUSTABLE), THE COMPRESSOR SHALL BE ENERGIZED AND STAGE ON MECHANICAL COOLING. ON A FALL IN SPACE TEMPERATURE THE REVERSE SHALL OCCUR.

HEATING:

WHEN THE SYSTEM IS INDEXED TO HEATING AND THE SPACE TEMPERATURE FALLS BELOW THE SPACE TEMPERATURE SETPOINT, INITIALLY SET AT 70°F (ADJUSTABLE), THE REVERSING VALVE SHALL BE ENERGIZED AND STAGE ON HEATING. ON A CONTINUED FALL IN SPACE TEMPERATURE THE ELECTRICAL HEATING COIL SHALL BE ENERGIZED. ON A RISE IN SPACE TEMPERATURE THE REVERSE SHALL OCCUR.

ADDITIONAL CONTROLS SHALL INCLUDE DEFROST CYCLE AND EMERGENCY HEAT.

DUCTLESS SPLIT-SYSTEM AIR HANDLING UNIT SCHEDULE

TAG	TYPE	CFM	OA (CFM)	TOTAL INPUT (W)	COOLING			ELECTRICAL DATA				NOTES
					TOTAL CAPACITY (BTU/HR)	SENSIBLE CAPACITY (BTU/HR)	EAT DB/WB (DEG. F)	VOLTS	PHASE	MCA	MFS	
DS-1	WALL MOUNTED	635	-	1960	24,000	18,000	80/67	208	1	1.0	15	1,2,3,4,5,6
DS-2	WALL MOUNTED	635	-	1960	24,000	18,000	80/67	208	1	1.0	15	1,2,3,4,5,6

NOTES:

1. PROVIDE UNIT WITH MANUFACTURER'S INTEGRAL CONDENSATE PUMP AND DISCONNECT SWITCH.
2. UNIT MUST OPERATE DOWN TO 0° F.
3. INDOOR UNIT POWER SUPPLIED FROM OUTDOOR UNIT. COORDINATE WITH ELECTRICAL CONTRACTOR.
4. CAPACITY BASED ON INDOOR CONDITIONS OF 80° FDB; 67° FWB. MINIMUM 21.4 SEER UNIT BASED ON ARI.
5. PROVIDE WITH MANUFACTURER'S LINE SET AND PROGRAMMABLE WALL MOUNTED WIRED CONTROLLER.
6. CFM BASED ON LOW SPEED.

SEQUENCE OF OPERATION

DS-1/2 AND CU-1/2 CONTROL

COOLING:

ON A CALL FOR COOLING THE MANUFACTURER'S WALL MOUNTED CONTROLLER SHALL STAGE ON MECHANICAL COOLING (SETPOINT @ 72° F). ON A FALL IN SPACE TEMPERATURE, THE REVERSE SHALL OCCUR.



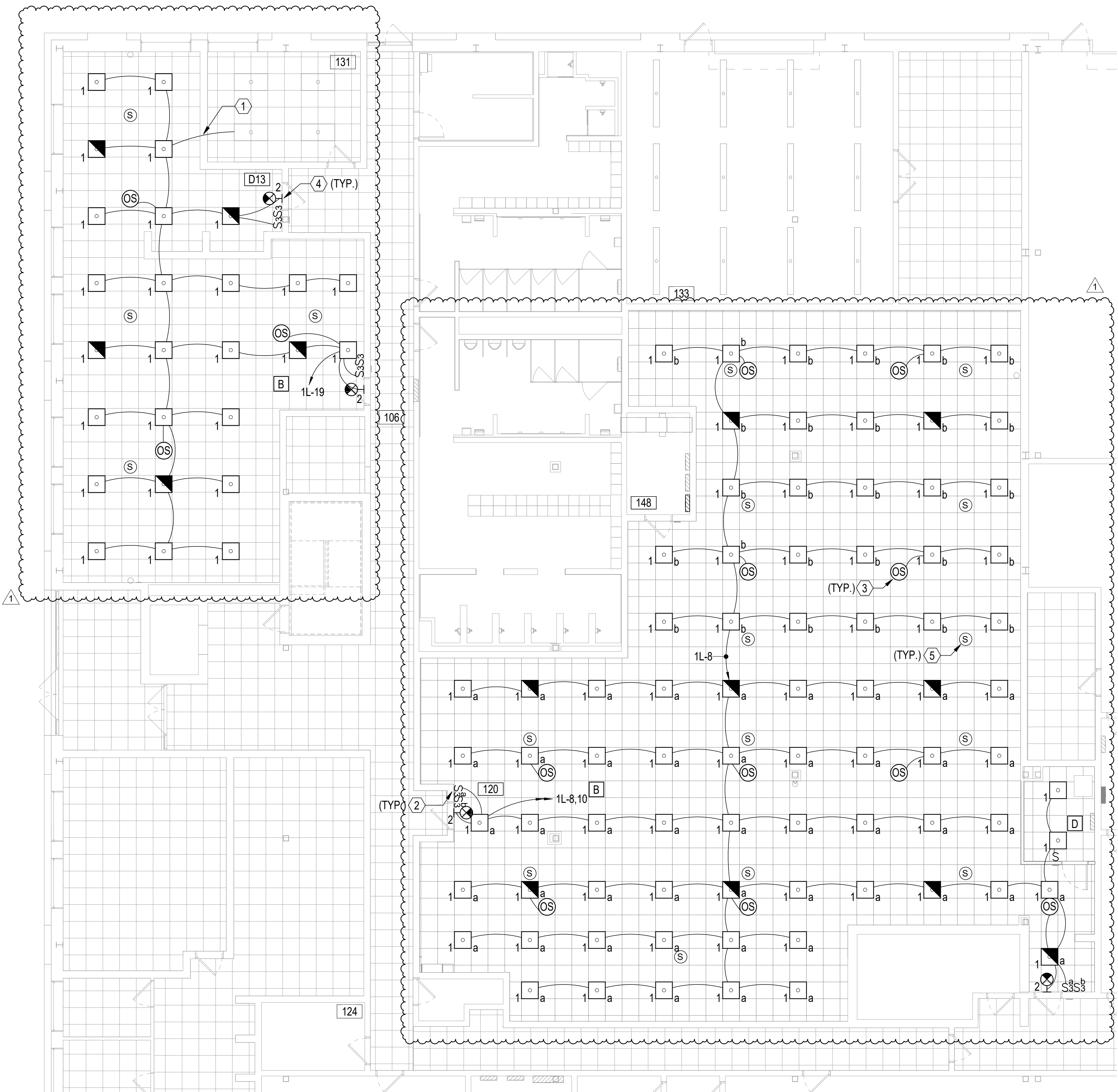
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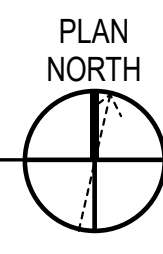
C

D



A1 BLDG CEP-209 - PARTIAL LIGHTING PLAN - NEW WORK
1/8" = 1'-0"

1/8" = 1'-0" 0 5' 10' 15' 30'



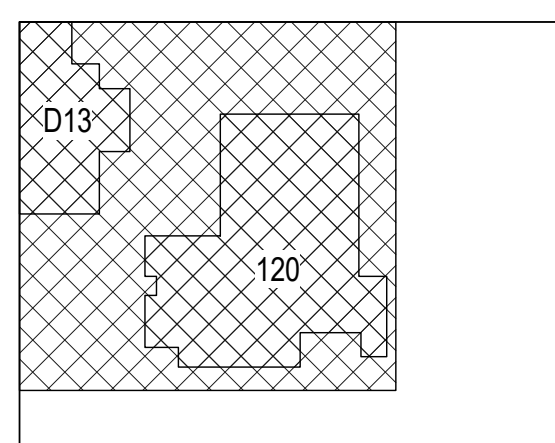
GENERAL NOTES

- A SEE SHEET E-001 FOR LEGEND, ABBREVIATIONS, AND GENERAL NOTES.
- B SEE SHEET EP401 FOR LOCATION OF MAIN ELECTRICAL ROOM IN BUILDING CEP-209. SEE SHEET EP401 FOR ENLARGED ELECTRICAL ROOM LAYOUTS.
- C SEE SHEET KEYPLAN FOR ELECTRICAL ROOM LOCATIONS IN BUILDING CEP-200. LIGHTING CIRCUITS FED FROM PANEL HA IN ELECTRICAL ROOM 084.
- D COORDINATE INSTALLATION OF COUNTERTOP RECEPTACLES WITH FURNITURE PLANS PRIOR TO BEGINNING WORK.
- E REFERENCE TT SERIES DRAWINGS FOR TELECOMMUNICATIONS SYSTEMS.
- F POWER AND DATA/TELEPHONE MAY BE CO-LOCATED IN SAME POWER POLE. FOR SYSTEMS FURNITURE WHIPS, PROVIDE SEPARATE OUTLET BOXES AND WHIPS BETWEEN POWER AND DATA/TELEPHONE. LOCATE BOXES ADJACENT TO EACH OTHER.

KEYNOTES

- 1 RECONNECT EXISTING FIXTURES TO LIGHTING CIRCUIT. PROVIDE WIRE AND CONDUIT TO EXTEND CIRCUIT TO RECONNECT FIXTURES AS REQUIRED.
- 2 PROVIDE 3 WAY DIMMER SWITCH CONTROLLING LIGHTING ZONES AS INDICATED ON PLAN.
- 3 APPROXIMATE LOCATION OF CEILING MOUNTED OCCUPANCY SENSOR SHOWN. PROVIDE AMOUNT AND LOCATION OF OCCUPANCY SENSORS AS REQUIRED FOR FULL COVERAGE OF SPACE PER MANUFACTURERS REQUIREMENTS.
- 4 CONNECT EXIT SIGNS TO LIGHTING CIRCUIT AHEAD OF SWITCHING.
- 5 SEE SHEET TT102 FOR CEILING SPEAKER DETAILS.

KEY PLAN



BUILDING CEP-209



APPROVED		DATE	APPR
FOR COMMANDER NAVFAC		4/25/25	
ACTIVITY		SYM	DESCRIPTION
NAVSEA		1	Design Revision 1
SATISFACTORY TO DATE		5/6/24	
DES	BMN	DRW	BMN
CHK	FVP		
PMIDM	SARAH GILPIN, R.A.		
BRANCH MANAGER	JEFF MORTON, P.E.		
CHIEF ENGINEER	JOHN MILLER, P.E.		
FIRE PROTECTION			
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC			
NAVAL STATION - NORFOLK, VA			
NORFOLK, VIRGINIA			
BUILDINGS CEP-200 & CEP-209			
FMB OFFICE RENOVATION			
BLDG CEP-209 - PARTIAL LIGHTING PLAN - NEW WORK			
SCALE: AS NOTED			
PROJECT NO.: 1637041			
CONSTR. CONTR. NO. N/A			
NAVFAC DRAWING NO. 12,902,291			
SHEET 56 OF 78			
EL102			
DRAWING REVISION: 25 AUGUST 2020			

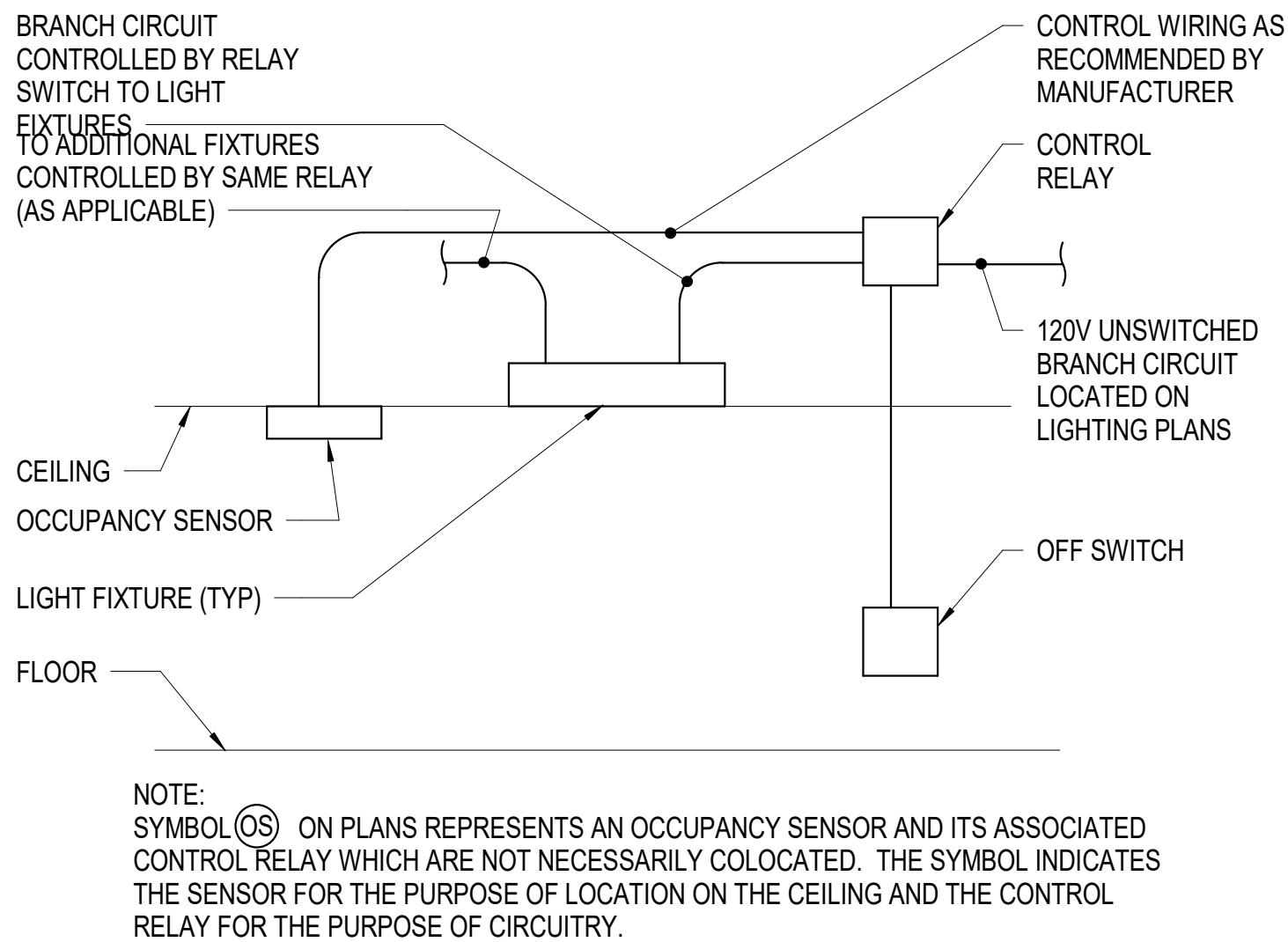
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EL501

A

4 OCCUPANCY SENSOR CONNECTION DIAGRAM

NOT TO SCALE



LIGHTING CONTROL SEQUENCE SCHEDULE		
SEQUENCE ID	LIGHTING SEQUENCE	ROOM LIGHTING SEQUENCE OF OPERATIONS
A	MANUAL ON/VACANCY OFF	OCCUPANT MANUALLY TURNS ON LIGHTS UPON ENTERING SPACE. UPON SENSING VACANCY AFTER 30 MINUTES, LIGHTS TURN OFF.
B	OCCUPANCY ON/VACANCY OFF	OCCUPANT ENTERS SPACE AND LIGHTS AUTOMATICALLY TURN ON. UPON VACANCY AFTER 30 MINUTES, LIGHTS TURN OFF.
C	OCCUPANCY 50% ON/MANUAL ADJUST/VACANCY OFF	OCCUPANT ENTERS SPACE AND LIGHTS AUTOMATICALLY TURN ON TO 50% OF FULL OUPUT. OCCUPANT CAN MANUALLY ADJUST LIGHT LEVELS. UPON VACANCY AFTER 30 MINUTES, LIGHTS TURN OFF
D	MANUAL ON/MANUAL OFF	OCCUPANT MANUALLY TURNS ON LIGHTS UPON ENTERING SPACE AND MANUALLY TURNS OFF LIGHTS WHEN LEAVING SPACE.

NOTE: THIS SKETCH IS A NON-PROPRIETARY GRAPHIC REPRESENTATION OF A LUMINAIRE THAT MAY MEET THE SPECIFICATION REQUIREMENTS. IT IS NOT INTENDED TO INDICATE A CERTAIN MANUFACTURER OR PREFERENCE.

LUMINAIRE REQUIREMENTS:

- HOUSING - HEAVY GAUGE COLD ROLLED STEEL OR DIE CAST ALUMINUM. SIZE SHOWN AS INDICATED IN LUMINAIRE SCHEDULE.
- OPTICS - FROSTED ACRYLIC OR POLYCARBONATE LENS WITH DIE FORMED COLD ROLLED SHEET STEEL REFLECTORS.
- LIGHT SOURCE - SOLID STATE LEDS, 3500K CCT UON, MINIMUM 80 CRI UON, AND MINIMUM EFFICACY OF 100 LUMENS/WATT UON. INITIAL LUMEN OUTPUT AS INDICATED IN LUMINAIRE SCHEDULE.
- DRIVER - REPLACEABLE, INTEGRAL, HIGH-EFFICIENCY DIMMABLE DRIVER WITH MINIMUM 0.9 PF, OPERATING VOLTAGE OF 120-277V, THERMAL MANAGEMENT, AND < 20% THD. ON/OFF CONTROL AND FULLY DIMMABLE DOWN TO 10% MINIMUM OR AS INDICATED IN LUMINAIRE SCHEDULE.
- CERTIFICATION - UL LISTED FOR DRY OR DAMP LOCATION, ROHS COMPLIANT, DLC QUALIFIED. COMPLIES WITH IES LM79, LM80 AND TM21 TESTING STANDARDS.
- MOUNTING - RECESSED IN HARD OR ACOUSTICAL TILE CEILING.
- OPTIONS - EMERGENCY BATTERY BACK-UP, INTEGRAL OCCUPANCY/VACANCY SENSOR, VARIOUS SIZE AND OUTPUT OPTIONS, SURFACE-MOUNTING KIT.

DIRECT/INDIRECT LED LUMINAIRE

REVISED: NOVEMBER 2020 LIGHTING PLATE: NL-1

NOTE: THIS SKETCH IS A NON-PROPRIETARY GRAPHIC REPRESENTATION OF A LUMINAIRE THAT MAY MEET THE SPECIFICATION REQUIREMENTS. IT IS NOT INTENDED TO INDICATE A CERTAIN MANUFACTURER OR PREFERENCE.

LUMINAIRE REQUIREMENTS:

- HOUSING - EXTRUDED ALUMINUM WITH CLEAR ACRYLIC EDGE-LIT PANEL.
- LIGHT SOURCE - SOLID STATE LEDS.
- DRIVER - INTEGRAL, HIGH-EFFICIENCY DRIVER WITH MINIMUM 0.9 PF, OPERATING VOLTAGE OF 120-277V, THERMAL MANAGEMENT, AND < 20% THD.
- CERTIFICATION - NFPA 101, UL LISTED FOR DAMP OR WET LOCATION, AND ROHS COMPLIANT.
- MOUNTING - SURFACE MOUNTED ON CEILING AND/OR WALL.
- OPTIONS - RED OR GREEN LETTERING, ONE- OR TWO-SIDED. BATTERY BACKUP.

EDGE-LIT EXIT SIGN

REVISED: NOVEMBER 2020 LIGHTING PLATE: NL-27

NOTE: THIS SKETCH IS A NON-PROPRIETARY GRAPHIC REPRESENTATION OF A LUMINAIRE THAT MAY MEET THE SPECIFICATION REQUIREMENTS. IT IS NOT INTENDED TO INDICATE A CERTAIN MANUFACTURER OR PREFERENCE.

LUMINAIRE REQUIREMENTS:

- HOUSING - DIE-CAST OR EXTRUDED ALUMINUM WITH INTEGRAL PASSIVE COOLING MECHANISM. HEAT SINK INCORPORATED DIRECTLY INTO HOUSING OR DRIVER COMPARTMENT.
- OPTICS - PRECISION MOLDED ACRYLIC LENS WITH TYPE II, III, OR IV DISTRIBUTIONS. BUG UPLIGHT RATING OF U0, WITH GLARE RATING AS DETERMINED BY LIGHTING ZONE INSTALLED.
- LIGHT SOURCE - SOLID STATE LEDS, 3000K CCT UON, MINIMUM 70 CRI UON, AND MINIMUM EFFICACY OF 80 LUMENS/WATT UON. INITIAL LUMEN OUTPUT AS INDICATED IN LUMINAIRE SCHEDULE.
- DRIVER - REPLACEABLE, INTEGRAL, HIGH-EFFICIENCY DIMMABLE DRIVER WITH MINIMUM 0.9 PF, OPERATING VOLTAGE OF 120-277V, THERMAL MANAGEMENT, AND < 20% THD. ON-OFF CONTROL AND FULLY DIMMABLE DOWN TO 10% MINIMUM OR AS INDICATED IN LUMINAIRE SCHEDULE.
- CERTIFICATION - UL LISTED FOR WET LOCATION, ROHS COMPLIANT. COMPLIES WITH IES LM79, LM80 AND TM21 TESTING STANDARDS.
- MOUNTING - SURFACE MOUNTED WITH STAINLESS STEEL MOUNTING HARDWARE.
- OPTIONS - VARIOUS LIGHT DISTRIBUTIONS. INTEGRAL MOTION SENSOR, PHOTOCCELL, BATTERY BACK-UP.

LED WALL PACK

REVISED: NOVEMBER 2020 LUMINAIRE PLATE: XL-10

EL501

DRAWING REVISION: 25 AUGUST 2020

SHEET 57 OF 78

NAVJAC DRAWING NO. 12.902.292

CONSTR. CONTR. NO. N/A

PROJECT NO.: 1637041

SCALE: AS NOTED

DEPARTMENT OF THE NAVY

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND

MID-ATLANTIC

NAVAL STATION NORFOLK

NORFOLK, VIRGINIA

BUILDINGS CEP-200 & CEP-209

FMB OFFICE RENOVATION

LIGHTING DIAGRAMS AND DETAILS

NAVSEA

ACTIVITY

FOR COMMANDER NAVJAC

APPROVED

DATE 5/6/24

SATISFACTORY TO DATE

DES BMN DRW BMN CHK Checker

PMIDM SARAH GILPIN, R.A.

BRANCH MANAGER JEFF MORTON, P.E.

CHIEF ENGINEER JOHN MILLER, P.E.

FIRE PROTECTION

NAVSEA

APPROVED

DATE 5/6/24

SATISFACTORY TO DATE

DES BMN DRW BMN CHK Checker

PMIDM SARAH GILPIN, R.A.

BRANCH MANAGER JEFF MORTON, P.E.

CHIEF ENGINEER JOHN MILLER, P.E.

FIRE PROTECTION

NAVSEA

ACTIVITY

FOR COMMANDER NAVJAC

APPROVED

DATE 5/6/24

SATISFACTORY TO DATE

DES BMN DRW BMN CHK Checker

PMIDM SARAH GILPIN, R.A.

BRANCH MANAGER JEFF MORTON, P.E.

CHIEF ENGINEER JOHN MILLER, P.E.

FIRE PROTECTION

NAVSEA

ACTIVITY

FOR COMMANDER NAVJAC

APPROVED

DATE 5/6/24

SATISFACTORY TO DATE

DES BMN DRW BMN CHK Checker

PMIDM SARAH GILPIN, R.A.

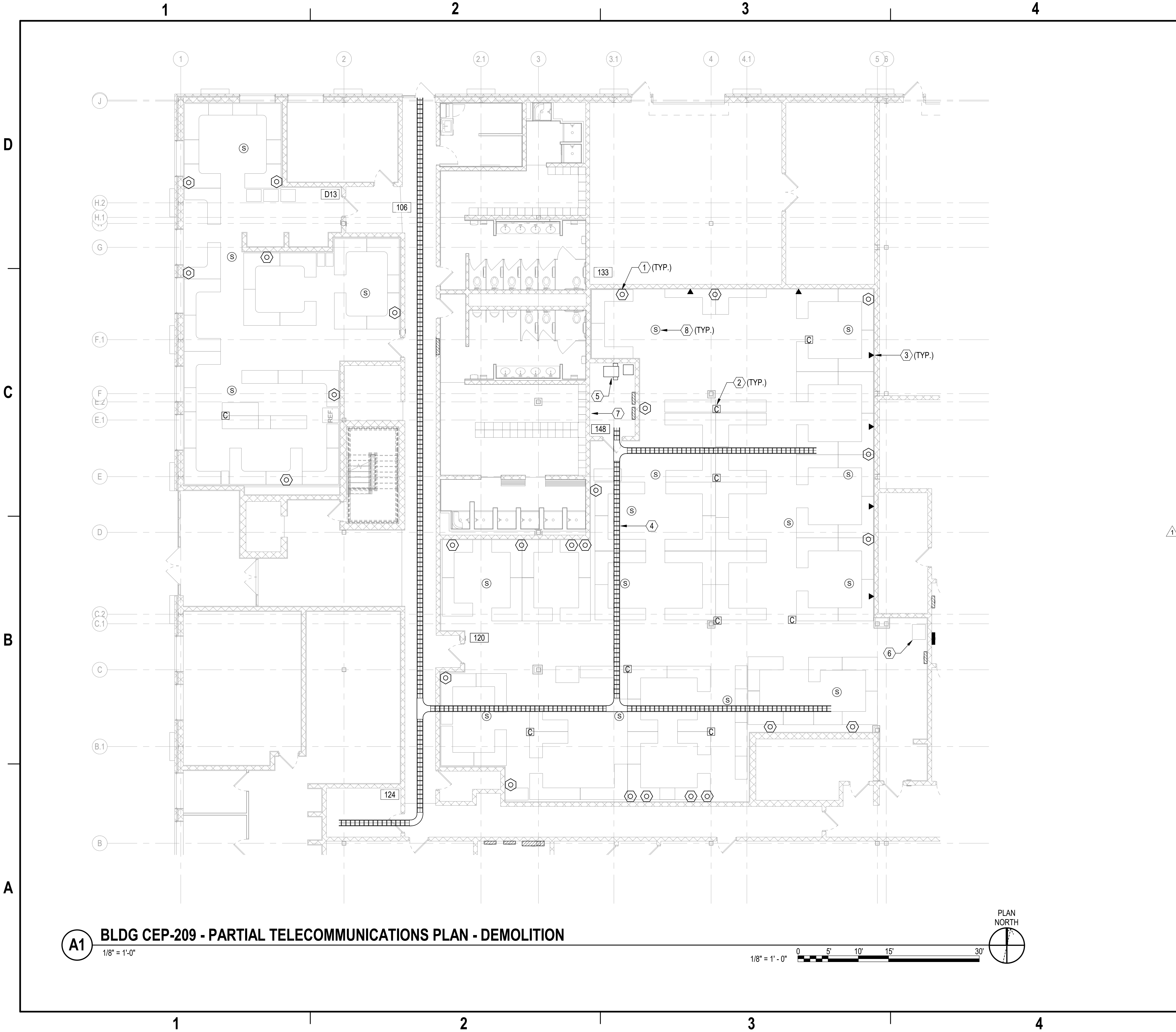
BRANCH MANAGER JEFF MORTON, P.E.

CHIEF ENGINEER JOHN MILLER, P.E.

FIRE PROTECTION

DRAWFORM REVISION: 25 AUGUST 2020

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A1 BLDG CEP-209 - PARTIAL TELECOMMUNICATIONS PLAN - DEMOLITION
1/8" = 1'-0"

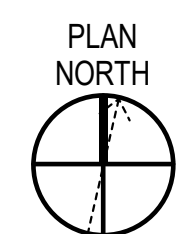
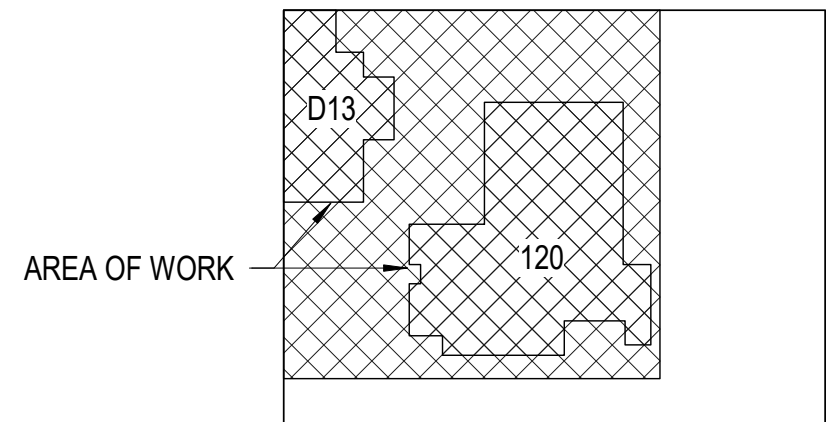
GENERAL NOTES

- 1 REFER TO SHEET T-001 FOR LEGEND, ABBREVIATIONS, AND GENERAL NOTES.
- 2 TELEPHONE EQUIPMENT IS LOCATED IN ELECTRICAL RM 084 IN CEP-200 AND LAN RM 148 IN CEP-209. DATA EQUIPMENT IS LOCATED IN LAN RM 148 IN CEP-209. COORDINATE EQUIPMENT LOCATION WITH ACTIVITY IN CEP-200, ASSUME 300' DISTANCE TO EQUIPMENT FOR BIDDING PURPOSES.
- 3 PROVIDE TEMPORARY PATCH PANEL FOR CAT5E CABLE TESTING. PROVIDE FINAL TEST REPORTS TO CONSTRUCTION OFFICE UPON COMPLETION. FINAL PUNCH DOWN TO DATA RACK TO BE MADE BY OTHER. PROVIDE 10' SLACK, LABEL, AND COIL ALL CABLES ABOVE EQUIPMENT RACK.
- 4 REUSE EXISTING CABLE TRAY FOR CABLE RUNS WHERE AVAILABLE. PROVIDE 1" CONDUIT OR J-HOOKS FOR ALL OTHER CABLE RUNS AND BETWEEN CABLE TRAY AND PUNCH DOWN LOCATION.

KEYNOTES

- 1 DISCONNECT EXISTING SYSTEMS FURNITURE TELECOMMUNICATIONS CONNECTION FROM SYSTEMS FURNITURE. DEMOLISH DATA AND TELEPHONE CABLING BACK TO SOURCE. DEMOLISH ASSOCIATED SURFACE MOUNTED RACEWAY AND CONDUIT BACK TO NEAREST UPSTREAM JUNCTION BOX.
- 2 DISCONNECT AND DEMOLISH EXISTING POWER POLE FOR SYSTEMS FURNITURE, DATA AND TELEPHONE CABLING, AND CONDUIT BACK TO SOURCE.
- 3 DEMOLISH EXISTING SURFACE MOUNTED DATA PORT, DATA CABLING, AND SURFACE MOUNTED RACEWAY AND CONDUIT BACK TO SOURCE.
- 4 PRESERVE EXISTING CABLE TRAY DURING CONSTRUCTION.
- 5 EXISTING DATA RACK TO REMAIN.
- 6 EXISTING DATA RACK TO REMAIN IN OPERATION DURING CONSTRUCTION. PROVIDE DUST COVER AND PROTECT DATA RACK DURING CONSTRUCTION.
- 7 EXISTING TELEPHONE/DATA EQUIPMENT. SEE SHEET TT502 FOR EQUIPMENT ELEVATION.
- 8 DISCONNECT EXISTING SPEAKER AND PRESERVE FOR USE IN NEW WORK. DEMOLISH (2) #16 TWISTED PAIR IN 1/2" C BACK TO JUNCTION BOXES IN RM 133 AND CORRIDOR 106. PRESERVE SPEAKER CIRCUITS FOR USE IN NEW WORK.

KEY PLAN



DATE 4/25/25		APPR
SYM 1		DESCRIPTION Design Revision 1
SEAL		
APPROVED FOR COMMANDER NAVFAC 5-9-25		
ACTIVITY NAVSEA		
SATISFACTORY TO DATE 5/6/24		
DES BMN	DRW BMN	CHK FVP
PMOM	SARAH GILPIN, R.A.	
BRANCH MANAGER	JEFF MORTON, P.E.	
CHIEF ENGINEER	JOHN MILLER, P.E.	
FIRE PROTECTION		
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND NAVAL STATION - NORFOLK, VA NORFOLK, VIRGINIA		
BUILDINGS CEP-200 & CEP-209 FMB OFFICE RENOVATION		
BLDG CEP-209 - PARTIAL TELECOMMUNICATIONS PLAN - DEMOLITION		
SCALE: AS NOTED		
EPROJCT NO.: 1637041		
CONSTR. CONTR. NO. N/A		
NAVFAC DRAWING NO. 12,902,307		
SHEET 72 OF 78		
TD102		
DRAWING REVISION: 25 AUGUST 2020		

SECTION 01 32 17.00 20

COST-LOADED NETWORK ANALYSIS SCHEDULES (NAS)
05/18, CHG 3: 08/20

PART 1 GENERAL

1.1 DEFINITIONS

The cost-loaded Network Analysis Schedule (NAS) is a tool to manage the project, both for Contractor and Government activities. The NAS is also used to report progress, evaluate time extensions, and provide the basis for progress payments.

For consistency, when scheduling software terminology is used in this section, the terms in Primavera's scheduling programs are used.

1.2 SCHEDULE REQUIREMENTS PRIOR TO THE START OF WORK

1.2.1 Preliminary Scheduling Meeting

Before preparation of the Project Baseline Schedule, and prior to the start of work, meet with the Contracting Officer to discuss the proposed schedule and the requirements of this section.

1.2.2 Project Baseline Schedule

Submit the Baseline NAS at the preconstruction meeting. Government review comments on the Contractor's schedule(s) do not relieve the Contractor from compliance with the requirements of the Contract Documents. Only bonds may be paid prior to acceptance of the Baseline NAS. The acceptance of a Baseline NAS is a condition precedent to:

- a. The Contractor starting demolition work or construction stage(s) of the contract.
- b. Processing Contractor's invoices(s) for any items other than bonds.
- c. Review of any schedule updates.

Submittal of the Baseline NAS is the Contractor's certification that the submitted schedule meets the requirements of the Contract Documents and represents the Contractor's plan on how the work will be accomplished, and accurately reflects the work that has been accomplished and how it was sequenced (as-built logic).

1.3 THREE-WEEK LOOK AHEAD SCHEDULE

1.3.1 Weekly CQC Coordination and Production Meeting

Deliver electronic file of 3-Week Look Ahead Schedule to the Contracting Office at least 24 hours prior to the weekly scheduled CQC Coordination and Production Meeting. Contractor is required to provide all attendees at the CQC Coordination and Production Meeting with a hard copy of the 3-Week Look Ahead Schedule.

1.3.2 Look Ahead Schedule Requirements

Prepare and issue a 3-Week Look Ahead schedule to provide a more detailed day-to-day plan of upcoming work identified on the Project Network Analysis Schedule. Requirements include:

- a. For each Look Ahead schedule activity, identify parent NAS activity number(s). The parent NAS activity is the activity in the NAS that would incorporate the Look Ahead schedule activity requirement and or scope of work.
- b. Update schedule each week to show the planned work for the current and following two-week period. Also include previous week, as-built work, showing actual start and finish dates.
- c. Include upcoming outages, closures, preparatory meetings, and initial meetings, testing and inspections.
- d. Clearly identify longest path activities on the Three-Week Look Ahead Schedule. Include a key or legend that distinguishes longest path activities. Include all Longest Path activity NAS start/finish dates exceeded and/or occurring during this period.
- e. The detail work plans are to be bar chart type schedules, derived from but maintained separately from the Project NAS on an electronic spreadsheet program and printed on 11 by 17 inch sheets as directed by the Contracting Officer.
- f. Activities must not exceed 5 working days in duration and have sufficient level of detail to assign crews, tools and equipment required to complete the work.

1.4 MONTHLY NETWORK ANALYSIS

Submittal of Monthly NAS is the Contractor's certification that the submitted schedule meets the requirements of the Contract Documents and represents the Contractor's plan on how the work will be accomplished. Provide all items listed in paragraph REQUIRED TABULAR REPORTS AND NATIVE P6 XER FILES with the monthly NAS submittal.

1.4.1 Monthly Network Analysis Updates

Meet with Government representative(s) at monthly intervals to review and agree on the information presented in the updated project schedule. The submission of an accepted, updated schedule to the Government is a condition precedent to the processing of the Contractor's invoice. Submit an acceptable, updated schedule to the Government regardless of whether a Contractor's invoice is submitted for the given period. The Contractor and Government must consent to agree on unit quantities of work completed evaluating to a percentage of payment for each activity progressed during the update period. Monthly update schedules must incorporate as-built events as they occurred and provide ongoing stats of anticipated finish dates. As-built events must correspond to contemporaneous records including but not limited to submittals, daily production reports and quality control reports.

Provide the following with each Schedule submittal:

- a. Time-Scaled Logic Diagram
- b. Reports listed in paragraph REQUIRED TABULAR REPORTS.
- c. Data disks containing the project schedule. Include the back-up native .xer program files.

1.5 CORRESPONDENCE AND TEST REPORTS

Reference Schedule activity IDs that are being addressed in each correspondence (e.g., letters, Requests for Information (RFIs), e-mails, meeting minute items, Production and QC Daily Reports, material delivery tickets, photographs) and test report (e.g., concrete, soil compaction, weld, pressure).

1.6 ADDITIONAL SCHEDULING REQUIREMENTS

Other specification sections may include additional scheduling requirements, including systems to be inspected, tested and commissioned, and submittal procedures. Those schedule requirements must be incorporated into the NAS schedule.

1.7 SUBMITTALS

Government approval/acceptance is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Baseline NAS; G

Designated Project Scheduler; G

SD-07 Certificates

Three-Week Look Ahead Schedule; G

Monthly Network Analysis Updates; G

1.8 SOFTWARE

Prepare and maintain project schedules using Primavera P6 software in a version compatible with Government's current version. Importing data into P6 using data conversion techniques or third party software is cause for rejection of the submitted schedule. Schedules with Performing Organizational Breakdown Structure (POBS) data is cause for rejection.

1.9 DESIGNATED PROJECT SCHEDULER

Within 30 calendar days of contract award, submit to the Contracting Officer for approval an individual who will serve as the Designated Project Scheduler. Include a copy of the candidate's resume with qualifications. The Contracting Officer may remove the Designated Project Scheduler, and require replacement, if the scheduler does not effectively

fulfill their duties in accordance with the contract requirements. Payment request will not be processed without an approved Designated Project Scheduler.

1.9.1 Qualifications

The Designated Project Scheduler must have prepared and maintained at least three previous construction schedules, of similar size and complexity to this contract, using Primavera P6.

1.9.2 Duties

Duties of the Designated Project Scheduler:

- a. Prepare Baseline NAS.
- b. Prepare monthly schedule updates.
- c. Prepare tabular reports.
- d. Prepare Time Impact Analysis (TIA) as necessary.
- e. Provide certification that NAS and TIA submittals conform to the contract requirements.
- f. Participate with the Prime Contractor and Government Representative in a monthly teleconference call or meeting at the job site in-person, and scheduled with sufficient time to support the Monthly Network Analysis Updates process, to discuss project status, schedule updates, critical activities, potential delays, and contract modifications impacting the schedule.

1.10 NETWORK SYSTEM FORMAT

Prepare the schedule in accordance with the following Primavera P6 settings and parameters. Deviation from these settings and parameters, without prior consent of the Contracting Officer, is cause for rejection of schedule submission.

1.10.1 Schedule Activity Properties and Level of Detail

1.10.1.1 Activity Identification and Organization

- a. Identify construction activities planned for the project and other activities that could impact project completion if delayed in the NAS.
- b. Each activity must have a unique name.
- c. Identify administrative type activity/milestones, including all pre-construction submittal and permit requirements prior to demolition or construction stage.
- d. Include times for procurement, Contractor quality control and construction, acceptance testing and training in the schedule.
- e. Include the Government approval time required for the submittals that require Government Approval prior to construction, as indicated in Section 01 33 00 SUBMITTAL PROCEDURES.

- f. Create separate activities for each Phase, Area, Floor Level and Location the activity is occurring.
- g. Do not use construction category activity to represent non-work type reference (e.g., Serial Letter, Request for Information) in NAS. Place Non-work reference within the P6 activity details notebook.

Activity categories included in the schedule are specified below.

1.10.1.2 Activity Logic

- a. With the exception of the Contract Award and Contract Completion Date (CCD) milestone activities, activity must not be open-ended; each activity must have at least one predecessor and at least one successor.
- b. Activities must not have open start or open finish (dangling) logic.
- c. Do not use lead or lag logic without Contracting Officer prior approval.
- d. Minimize redundant logic ties.
- e. Once an activity exists on the schedule it must not be deleted or renamed to change the scope of the activity and must not be removed from the schedule logic without approval from the Contracting Officer.
 - (1) While an activity cannot be deleted, where said activity is no longer applicable to the schedule, but must remain within the logic stream for historical record, change the activity original and remaining duration to zero and clearly label "(NO LONGER REQUIRED)" after the activity name. Actual finish date for activity that falls behind the data date. Redistribute accordingly any remaining budget associated with that activity, to other remaining appropriate activity.
 - (2) Document any such change in the activities' "Notebook," including a date and explanation for the change.
 - (3) The ID number for a "NO LONGER REQUIRED" activity must not be re-used for another activity.

1.10.1.3 Longest Path Activity Baseline Limitation

For P6 settings, critical activities are defined as being on the Longest Path. Longest Path (Critical) Activities must not make up more than 30 percent of all activity within the Construction Baseline Schedule.

1.10.1.4 Assigned Calendars

All NAS activity must be assigned calendars that reflect required and anticipated non-work days.

1.10.1.5 Activity Categories

1.10.1.5.1 Pre-construction Activities

Examples of pre-construction activities include, but are not limited to, bond approval, permits, pre-construction submittals and approvals. Include pre-construction activities that are required to be completed

prior to the Contractor starting the demolition or construction stage of work.

1.10.1.5.2 Procurement Activities

Examples of procurement activities include, but are not limited to: Material/equipment submittal preparation, submittal and approval of material/equipment; material/equipment fabrication and delivery, and material/equipment on-site. As a minimum, separate procurement activities must be provided for critical items, long lead items, items requiring Government approval and material/equipment procurement for which payment will be requested in advance of installation. Show each delivery with relationship tie to the Construction Activity specifically for the delivery.

1.10.1.5.3 Government Activities

Government and other agency activities that could impact progress must be clearly identified. Government activities include, but are not limited to; Government approved submittal reviews, Government conducted inspections/tests, environmental permit approvals by State regulators, utility outages, and delivery of Government Furnished Material/Equipment.

1.10.1.5.4 Construction Quality Management (CQM) Activities

The Preparatory and Initial Phase meetings for each Definable Feature of Work identified in the Contractor's Quality Control Plan must be included in the Three-Week Look Ahead Schedule. Preparatory and Initial phase meetings are not required in the NAS, but can be represented by a start milestone linked to successor parent Construction Activity. The Follow-up Phase must be represented by the Construction Activities themselves in the NAS.

1.10.1.5.5 Construction Activities

On-site construction activities must not have a duration in excess of 20 working days. Contractor activities must be driven by calendars that reflect Saturdays, Sundays and all Federal Holidays as non-work days, unless otherwise defined in this contract.

1.10.1.5.6 Turnover and Closeout Activities

Include activities or milestones for items on the NAVFAC Red Zone Checklist/POAM that are applicable to this project. As a minimum, include required Contractor testing, required Government acceptance inspections on equipment, Pre-Final Inspection, Punch List Completion, Final Inspection and Acceptance. Add an unconstrained start milestone for the initial NAVFAC Red Zone - Facility Turnover Planning Meeting at approximately 75 percent construction contract completion or six months prior to Contract Completion Date (CCD), whichever is sooner.

1.10.1.6 Contract Milestones and Constraints

1.10.1.6.1 NAVFAC Red Zone - Facility Turnover Planning Meeting Milestones

See paragraph TURNOVER AND CLOSEOUT ACTIVITIES above.

1.10.1.6.2 Substantial Completion Milestone

Include an unconstrained finish milestone on the schedule titled "Substantial Completion." Substantial Completion is defined as the point in time the Government would consider the project ready for beneficial occupancy wherein by mutual agreement of the Government and Contractor, Government use of the facility is allowed while construction access continues in order to complete remaining items (e.g., punch list and other close out submittals). Include a separate Substantial Completion Milestone for each phase if the contract requires construction to be completed in phases.

1.10.1.6.3 DD-1354 Finish Milestone

Add unconstrained finish milestone, titled "DD-1354" and scheduled 30 calendar days prior to Substantial Completion, whenever a Form DD-1354 is required in accordance with Section 01 78 00 CLOSEOUT SUBMITTALS.

1.10.1.6.4 Projected Completion Milestone

Include an unconstrained finish milestone on the schedule titled "Projected Completion." Projected Completion is defined as the point in time all contract requirements are complete and verified by the Government with a successful Final Inspection in accordance with Section 01 45 00.10 20 QUALITY CONTROL FOR MINOR CONSTRUCTION. This milestone must have the Contract Completion Date (CCD) milestone as its only successor.

1.10.1.6.5 Contract Completion Date (CCD) Milestone

Last schedule entry must be an unconstrained finish milestone titled "Contract Completion (CCD: DD-MM-YY)." DD-MM-YYYY is the current contract completion date at data date, day-month-year corresponding to P6 Must Finish By Date. NAS milestone updates of Project Completion finish date for longest path must reflect calculated float as positive or negative based on CCD. Calculation of schedule updates must be such that if the finish of the "Projected Completion" milestone falls after the contract completion date, then negative float is calculated on the longest path. If the finish of the "Projected Completion" milestone falls before the contract completion date, the float calculation must reflect positive float on the longest path.

1.10.1.7 Work Breakdown Structure & Activity Code

At a minimum, establish a Work Breakdown Structure (WBS) and provide activity codes identified as follows:

1.10.1.7.1 Work Breakdown Structure (WBS)

Group all activities and milestones within appropriate WBS categories including, at a minimum, the following:

a. Project Milestones:

- (1) Management Milestones
- (2) Project Administrative Meetings
- (3) Permits

- b. Pre-Construction Phase:
 - (1) Submittals and Reviews
 - (2) Procurement
 - (3) Mobilization
- c. Construction Phase: Create multiple sub-sections in accordance with project specific categories of work including in WBS descending order as follows:
 - (1) General Area
 - (a) Type of Work Item
 - 1. Location
- d. Project Closeout: Include activity items such as, but not limited to, Punchlist, Demobilization, O&M, As-built Drawings, Training, and As-built NAS.
- e. Modifications: Create sub-category of Conformed and Non-Conformed under Modification WBS. Create multiple sub-sections as the project progresses identified by issue and Fragnet placed in Conformed for modifications issued prior data date, or Non-Conformed for issues not modified to contract prior data date.
- f. Removed Activity: Activity is "removed" by remaining within logic sequence, eliminating duration and adding "(NO LONGER REQUIRED)" after Activity Name in Activity Table.

1.10.1.7.2 Responsibility Code

All activities in the project schedule must be identified with the resource for completing the task. Activities must not belong to more than one responsible party.

1.10.1.7.3 Activity Category Code

Provide user defined "CAT" codes for Project Level activity codes. Use the following codes:

- a. Assign "Procure" to Procurement type activity.
- b. Assign "Construct" Construction type activity.
- c. Assign "Close Out" to dedicated Commissioning, Testing & Close Out type activity.
- d. Assign "Other" to other activity not otherwise designated.

1.10.1.7.4 Drawing Code

Identify all activities in the project schedule with its respective Drawing Code. The Drawing Code is the Sheet Number on the primary project drawing which indicates work to be performed. If an activity does not have an applicable Drawing Code (e.g., Mobilize), the code must be "0000".

1.10.1.8 Adverse Weather Lost Work Days

Use the National Oceanic and Atmospheric Administration's (NOAA) Summary of Monthly Normals report to obtain the historical average number of days each month with precipitation, using a nominal 30-year, greater than 0.10 inch precipitation amount parameter, as indicated on the Station Report for the NOAA location closest to the project site as the basis for establishing a "Weather Calendar" showing the number of anticipated non-workdays for each month due to adverse weather, in addition to Saturdays, Sundays and all Federal Holidays as non-work days.

Assign the Weather Calendar to any activity that could be impacted by adverse weather. The Contracting Officer will issue a modification in accordance with the contract clauses, giving the Contractor a time only extension for the difference of days between the anticipated and actual adverse weather delay if the number of actual adverse weather delay days exceeds the number of days anticipated for the month in which the delay occurs and the adverse weather delayed activities are on the longest path to contract completion in the period when delay occurred. A lost workday due to weather conditions is defined as a day in which the Contractor cannot work at least 50 percent of the day on the impacted activity.

1.10.1.9 Cost Loading

The Project Network Analysis Schedule (NAS) must be cost-loaded and will provide the basis for progress payments.

1.10.1.9.1 Cost Loading Activities

Assign material and equipment costs, including their quantities, for which payment will be requested in advance of installation, to their respective procurement activity. Assign labor costs, including their quantities, for material and equipment paid for after installation to their respective construction activities. Include all typical mobilization costs dispersed over early construction activities. Costs for mobilization will not be paid as individual pay items with the exception of batch plant set-up, mobilization of dredging equipment or other similar labor-intensive situations. The value of commissioning, testing and closeout WBS section may not be less than 10percent of the total costs for procurement and construction activities. ALL activities assigned Government responsibility will have Zero Cost. No contractor cost should be assigned to an activity designated as a Government responsibility. Do not include field overhead positions as individual pay items. Evenly disperse overhead costs and profit to each activity over the duration of the project.

1.10.1.9.2 Partial Payment

Breakdown unit of measure and cost must be defined within P6 Activity Detail Expenses for partial payment of any cost loaded activity. Lump sum cost loaded activity will not be partially paid.

1.10.2 Schedule Software Settings and Restrictions

- a. Activity Constraints: Date/time constraint(s), other than those required by the contract, are not allowed unless accepted by the Contracting Officer. Identify any constraints proposed and provide an explanation for the purpose of the constraint in the Narrative Report as described in paragraph REQUIRED TABULAR REPORTS.

- b. Default Progress Data Disallowed: Actual Start and Actual Finish dates on the CPM schedule must match the dates on the Contractor Quality Control and Production Reports.
- c. Software Settings: Handle schedule calculations and Out-of_Sequence progress (if applicable) through Retained Logic, not Progress Override. Show all activity durations and float values in days. Show activity progress using Remaining Duration. Set default activity type to "Task Dependent".
- d. At a minimum, include the following settings and parameters in P6 Schedule preparation:
 - (1) General: Define or establish Calendars and Activity Codes at the "Project" level, not the "Global" level.
 - (2) Admin Drop-Down Menu, Admin Preferences, Time Periods Tab:
 - (a) Set time periods for P6 to 8.0 Hours/Day, 40.0 Hours/Week, 172.0 Hours/Month and 2000.0 Hours/Year.
 - (b) Use assigned calendar to specify the number of work hours for each time period: Must be checked.
 - (3) Admin Drop-Down Menu, Admin Preferences, Earned Value Tab:
 - (a) Earned Value Calculation: Use "Budgeted values with current dates".
 - (4) Project Level, Dates Tab:
 - (a) Set "Must Finish By" date to "Contract Completion Date", and set "Must Finish By" time to 05:00pm.
 - (5) Project Level, Defaults Tab:
 - (a) Duration Type: Set to "Fixed Duration & Units".
 - (b) Percent Complete Type: Set to "Physical".
 - (c) Activity Type: Set to "Task Dependent".
 - (d) Calendar: Set to "Standard 5 Day Workweek". Calendar must reflect Saturday, Sunday and all Federal holidays as non-work days. Alternative calendars may be used with Contracting Officer approval.
 - (6) Project Level, Calculations Tab:
 - (a) Default Price/Unit for activities without resource or role Price/Units: Set to "\$1/h".
 - (b) Activity percent complete based on activity steps: Must be Checked.
 - (c) Link Budget and At Completion for not started activities: Must be Checked.

(d) Reset Remaining Duration and Units to Original: Must be Selected.

(e) Subtract Actual from At Completion: Must be Selected.

(f) Recalculate Actual units and Cost when duration percent complete changes: Must be Checked.

(g) Update units when costs change on resource assignments: Must be Unchecked.

(h) Link Actual to Date and Actual This Period Units and Cost: Must be Checked.

(7) Project Level, Settings Tab:

(a) Define Critical Activities: Check "Longest Path".

(8) Work Breakdown Structure Level, Earned Value Tab:

(a) Technique for Computing Performance Percent Complete: "Activity percent complete" is selected.

(b) Technique for Computing Estimate to Complete (ETC): "PF = 1" is selected.

1.10.3 Required Tabular Reports and Native P6 XER Files

Include the following reports with the Baseline, Monthly Update and any other required schedule submittals:

a. Time Scaled Logic Schedule

Provide formatted 11 by 17-inch Time-scaled Logic Schedule in color and landscape-oriented with each schedule submittal. Clearly show activities on the longest path setting Gantt chart longest path activity bars to red. Group activities by WBS and sort by finish date in ascending order. Include the following information in column form for each activity and include accompanying Gantt chart:

(1) Activity ID

(2) Activity Name

(3) Original Duration

(4) Remaining duration

(5) Physical Percent Complete

(6) Start Date

(7) Finish Date

(8) Total Float

b. Previous Monthly Update Comparison Time Scaled Logic Schedule (Submit with all Monthly Update Schedule Submittals.)

Provide formatted 11 by 17-inch Time-scaled Logic Schedule in color and landscape-oriented with each monthly update schedule submittal. Clearly show activities on the current month longest path setting Gantt chart longest path activities bars to red. Show previous month activities as yellow bars and previous month milestones in yellow within Gantt chart. Sort by finish date in ascending order. Filter activities for longest path. Maintain and assign the accepted previous month update or the accepted baseline schedule for the first update submittal as the baseline and primary baseline in P6 before printing the schedule. Include the following information in column form for each activity and include accompanying Gantt chart:

- (1) Activity ID
 - (2) Activity Name
 - (3) Original Duration
 - (4) Current Month Remaining Duration
 - (5) Current Month Start Date
 - (6) Previous Month Update Start Date (BL Project Start)
 - (7) Start Date Delta between Current Month and Previous Month
(Variance - BL Project Start Date)
 - (8) Current Month Finish Date
 - (9) Previous Month Finish Date (BL Project Finish)
 - (10) Finish Date Delta between Current Month and Previous Month
(Variance - BL Project Start Date)
 - (11) Current Month Total Float
- c. P6 native XER file: Include the back-up native .xer program file compatible with the Government version of P6. Each native schedule file must have a unique file name to include project name and data date using (yyyy-mm-dd) convention. Each native schedule must have a unique Project ID and Project Name.
- d. Log Report: P6 Scheduling/Leveling Report.
- e. Narrative Report: Identify and justify:
- (1) Provide Project Summary Data in format below:
 - (a) Data Date _____
 - (b) Award Date: _____
 - (c) Original Project Duration: _____ days post Award Date
 - (d) Current Project Duration: _____ days post Award Date
 - (e) Time percent elapsed: _____ percent at data date
 - (f) Original CCD: _____

- (g) Current CCD: _____ (thru MOD _____)
 - (h) Anticipated CCD: _____ (____ calendar days early/late)
 - (i) Original Contract Value: \$_____
 - (j) Current Contract Value: \$_____
 - (k) Invoiced Amount: \$_____ (____ percent)
 - (l) Cost Growth: _____ percent
 - (m) Schedule Growth: _____ percent
 - (n) There are a total of _____ activities, _____ activities complete (____ percent), _____ activities in progress (____ percent), _____ activities not started (____ percent). Of the in progress and not started activities; _____ (____ percent) are on the longest path. The longest path has duration of _____ calendar days from data-date to anticipated project completion.
- (2) Progress made in each area of the project;
 - (3) Longest Path;
 - (4) Date/time constraint(s), other than those required by the contract
 - (5) Listing of all changes made between the previous schedule and current updated schedule include: added or deleted activities, original and remaining durations for activities that have not started, logic (sequence constraint lag/lead), milestones, planned sequence of operations, longest path, calendars or calendar assignments, and cost loading;
 - (6) Any decrease in previously reported activity Earned Amount;
 - (7) Pending items and status thereof, including permits, changes orders, and time extensions;
 - (8) Status of Contract Completion Date and interim milestones;
 - (9) Status of Projected Completion Milestone and account of difference in calendar days between previous update Projected Completion Milestone
 - (10) Current and anticipated delays listing Activity Names and IDs for impacted activities(describe cause of delay and corrective actions(s) and mitigation measures to minimize);
 - (11) Description of current and potential future schedule problem areas.
 - (12) Identification of any weather and restricted lost time as compared to anticipated weather for the month and anticipated restricted days for which the update is submitted. Impacts resulting from adverse weather must be documented in tabular form showing the calendar month (or billing period) with the days on which construction activity incurred Lost Work Days due to adverse

weather. In narrative form, describe the adverse weather cause such as precipitation measurement, temperature, wind or other influencing factors, and why work was impacted. Describe the construction activity(s) that was (were) scheduled, impacted.

Each entry in the narrative report must cite the respective Activity ID and Activity Name, the date and reason for the change, and description of the change.

- f. Earned Value Report: Derive from and correspond to P6 cost loaded schedule. List all activities having a budget amount cost loaded. Compile total earnings on the project from notice to proceed to current progress payment request. Show current budget, previous physical percent complete, to-date physical percent complete, previous earned value, to-date earned value, cost this period and cost to complete on the report for each activity.

1.11 CONTRACT MODIFICATION

1.11.1 Time Impact Analysis (TIA)

Submit a Time Impact Analysis with each cost and time proposal for a proposed change. TIA must illustrate the influence of each change or delay on the Contract Completion Date or milestones. No time extensions will be granted nor delay damages paid unless a delay occurs which consumes all available Project Float, impacts the longest path, and extends the Projected Completion beyond the Contract Completion Date.

- a. Each TIA must be in both narrative and schedule form. The narrative must define the scope and conditions of the change; provide start and finish dates of impact, successor and predecessor activity to impact period, responsible party; describe how it originated, and how it impacts the schedule. The schedule submission must consist of three native XER files:
 - (1) Fragnet used to define the scope of the changed condition
 - (2) Most recent accepted schedule update as of the time of the impact start date. Update this schedule to show all activity progress as of the time of the impact start date. The impact start date is identified as the time when an existing activity is impeded for either starting or finishing.
 - (3) The impacted schedule that has the fragnet inserted in the updated schedule and the schedule "run" so that the new completion date is determined.
- b. For claimed as-built project delay, the inserted fragnet TIA method must be modified to account for as-built events known to occur after the data date of schedule update used.
- c. All TIAs must include any mitigation, and must determine the apportionment of the overall delay assignable to each individual delay. Apportionment must provide identification of delay type and classification of delay by compensable and non-compensable events.

The associated narrative must clearly describe analysis methodology used, and the findings in a chronological listing beginning with the earliest delay event.

(1) Identify and classify types of delay defined as follows:

(a) Force majeure delay (e.g., weather delay): Any delay event caused by something or someone other than the Government or the Contractor, or the risk of which has not been assigned solely to the Government or the Contractor. If the force majeure delay is on the longest path, in absence of other types of concurrent delays, the Contractor is granted an extension of contract time, classified as a non-compensable event.

(b) A Contractor-delay: Any delay event caused by the Contractor, or the risk of which has been assigned solely to the Contractor. If the contractor-delay is on the longest path, in absence of other types of concurrent delays, Contractor is not granted extension of contract time, and classified as a non-compensable event. Where absent other types of delays, and having impact to project completion, Contractor must provide to Contracting Officer a Corrective Action Plan identifying plan to mitigate delay.

(c) A Government-delay: Any delay event caused by the Government, or the risk of which has been assigned solely to the Government. If the Government-delay is on the longest path, in absence of other types of concurrent delays, the Contractor is granted an extension of contract time, and classified as a compensable event.

(2) Functional concurrency must be used to analyze concurrent delays, where: separate delay issues delay project completion, do not necessarily occur at same time, rather occur within same monthly schedule update period at minimum, or within same as-built period under review. If a combination of functionally concurrent delay types occurs, it is considered Concurrent Delay, which is defined in the following combinations:

(a) Government-delay concurrent with contractor-delay: excusable time extension, classified non-compensable event.

(b) Government-delay concurrent with force majeure delay: excusable time extension, classified non-compensable event.

(c) Contractor-delay concurrent with force majeure delay: excusable time extension, classified non-compensable event.

(3) Pacing delay reacting to another delay (parent delay) equally or more critical than paced activity must be identified prior to pacing. Contracting Officer will notify Contractor prior to pacing. Contractor must notify Contracting Officer prior to pacing. Notification must include identification of parent delay issue, estimated parent delay time period, paced activity(s) identity, and pacing reason(s). Pacing Concurrency is defined as follows:

(a) Government-delay concurrent with contractor-pacing: excusable time extension, classified compensable event.

(b) Contractor-delay concurrent with Government-pacing:

inexcusable time extension, classified non-compensable event

- d. Submit electronic file containing the narrative and the source schedule files used in the time impact analysis.

1.12 PROJECT FLOAT

Project Float is the length of time between the Contractor's Projected Completion Milestone and the Contract Completion Date. Project Float available in the schedule will not be for the exclusive use of either the Government or the Contractor.

The use of Resource Leveling or other techniques used for the purpose of artificially adjusting activity durations to consume float and influence longest path is prohibited.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 05 52 00.00 27

NON PENETRATING PERMANENT PERSONNEL GUARDRAIL
07/17

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2015; Errata 1 2015; Errata 2 2016)
Structural Welding Code - Steel

ASTM INTERNATIONAL (ASTM)

ASTM A123/A123M (2015) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products

ASTM A153/A153M (2016) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware

ASTM A27/A27M (2013; R 2016) Standard Specification for Steel Castings, Carbon, for General Application

ASTM A283/A283M (2013) Standard Specification for Low and Intermediate Tensile Strength Carbon Steel Plates

ASTM A36/A36M (2014) Standard Specification for Carbon Structural Steel

ASTM A47/A47M (1999; R 2014) Standard Specification for Ferritic Malleable Iron Castings

ASTM A500/A500M (2013) Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes

ASTM A53/A53M (2012) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless

ASTM A575 (1996; E 2013; R 2013) Standard Specification for Steel Bars, Carbon, Merchant Quality, M-Grades

ASTM C780 (2016) Standard Test Method for Preconstruction and Construction Evaluation of Mortars for Plain and Reinforced Unit Masonry

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

NAAMM AMP 521

(2001) Pipe Railing Manual

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Fabrication Drawings; G

SD-03 Product Data

Steel Railings and Handrails; G

Anchor Systems; G

SD-05 Design Data

Design and Testing; G

SD-08 Manufacturer's Instructions

Manufacturer's Installation Instructions; G

PART 2 PRODUCTS

2.1 GENERAL REQUIREMENTS

Guardrail system must be permanent and must rest on top of the roofing system without attaching or penetrating the roofing. The system must be designed not to damage the roof during its service life.

The top of the rail must be 42 inches high above the walking level. Other structural members, including mid-rails, must be installed such that there are no openings in the guardrail system that are more than 19 inches high. Rails must be a minimum of 1 1/2 inch nominal diameter.

Railings must be either hot-dipped galvanized steel or aluminum.

Guardrails must be in accordance with OSHA 29 CFR 1926.502 and IMC 304.11.

2.2 DESIGN AND TESTING

Top rails shall withstand a minimum concentrated load of 200 pounds applied in any direction. Mid-rails, intermediate vertical members, and equivalent structural members must be capable of withstanding a force of 150 pounds applied in any downward or outward direction. Toe-boards must be capable of withstanding a force of 50 pounds applied in an outward direction. For design and testing, loads on top rails, and midrails not

to act concurrently.

Provide calculations or test results signed and stamped by an independent registered professional engineer that the guardrail system meets the load capacity requirements.

2.3 FABRICATION

Use materials of size and thicknesses indicated or, if not indicated, of required size and thickness to produce adequate strength and durability in finished product for intended use and other specified requirements. Work materials to dimensions indicated on approved detail drawings, using proven details of fabrication and support. Use type of materials indicated or specified for the various components of work.

Form exposed work true to line and level with accurate angles and surfaces and straight edges. Ensure all exposed edges are eased to a radius of approximately 1/32-inch. Bend metal corners to the smallest radius possible without causing grain separation or otherwise impairing the work.

Form exposed connections with hairline joints that are flush and smooth, using concealed fasteners wherever possible.

Provide anchoring devices and space as indicated and as required to provide adequate support for the intended use of the work. Provide removable railing sections as indicated.

Guardrail systems and Roof hatch safety railing systems must be surfaced to prevent injury from punctures or lacerations, and to prevent snagging of clothing.

2.3.1 Fabrication Drawings

Verify field measurements prior to submitting Fabrication drawings.

Provide railings detail plans, elevations, and sections at not less than 1-inch to 1-foot. Provide details of sections and connections at not less than 3-inches to 1-foot.

Indicate shapes, sizes, connections, size and type of fasteners, anchors, and accessories.

2.4 COMPONENTS

2.4.1 Steel Railings And Handrails

NAAMM AMP 521, provide the same size rail and post. Provide pipe collars of the same material and finish as the handrail and posts.

Weld corners and seams continuously and in accordance with the recommendations of AWS D1.1/D1.1M. Grind exposed welds smooth and flush to match and blend with adjoining surfaces.

Provide steel handrails, steel pipe conforming to ASTM A53/A53M or structural tubing conforming to ASTM A500/A500M, Grade A or B of equivalent strength. Provide steel railings of 1-1/2-inches nominal size, hot-dip galvanized and shop painted.

Provide toe-plates between railing posts where indicated, and consisting

minimally of 1/8-inch steel flat bars not less than 6-inches high.

2.4.1.1 Structural Steel Plates, Shapes And Bars

Provide structural-size shapes and plates, except plates to be bent or cold-formed, conforming to ASTM A36/A36M, unless otherwise noted.

Provide steel plates, to be bent or cold-formed, conforming to ASTM A283/A283M, Grade C.

Provide steel bars and bar-size shapes conforming to ASTM A36/A36M, unless otherwise noted.

2.4.1.2 Structural Steel Tubing

Provide structural steel tubing, hot-formed, welded or seamless, conforming to ASTM A500/A500M, Grade B, unless otherwise noted.

2.4.1.3 Hot-Rolled Carbon Steel Bars

Provide bars and bar-size shapes conforming to ASTM A575, grade as selected by the fabricator.

2.4.1.4 Steel Pipe

Provide pipe conforming to ASTM A53/A53M, type as selected, Grade B; primed finish, unless galvanizing is required; standard weight (Schedule 40).

2.4.1.5 Castings

Provide steel castings conforming to ASTM A27/A27M and iron castings conforming to ASTM A47/A47M.

2.4.1.6 Fasteners

Provide fasteners conforming to

Provide galvanized fasteners in accordance with ASTM A153/A153M. Select fasteners for the type, grade, and class required for the installation of railing items.

2.4.1.7 Welding

All steel welding work must comply with AWS D1.1/D1.1M.

2.4.1.8 Galvanizing

Hot dip galvanize railings, including pipe, fittings, brackets, fasteners, and other ferrous metal components after fabrication as much as practicable. Provide hot dipped galvanized steelwork in accordance with ASTM A123/A123M and ASTM A153/A153M.

Repair damage to galvanized coatings using ASTM C780 zinc rich paint for galvanizing damaged by handling, transporting, cutting, welding, or bolting. Do not heat surfaces to which repair paint has been applied. Zinc rich compound shall be 95% metallic zinc, by weight in dried film, conforming to Federal Specification DoD-P-21035A and meeting Mil Spec MIL-P-26915A.

2.4.2 Anchor Systems

Anchorage for the guardrail system and roof hatch safety railing system must not penetrate or attach to the roofing. Provide an adequate anchor system to meet the load requirements.

Provide rubber pads or other means of protecting the roofing surface and provide adequate frictional resistance to forces on the guardrail.

PART 3 EXECUTION

3.1 INSTALLATION

3.1.1 Manufacturer's Installation Instructions

Submit manufacturer's installation instructions, including diagrams, for the complete installation of the railings, self closing gates, anchorages, and accessories.

3.1.2 Touchup Galvanizing and Painting

Immediately after installation, clean field welds, bolted connections, abraded or exposed areas of the galvanizing or shop paint. For galvanized areas, provide two coats of galvanizing repair paint with 2 mil thickness per coat, for a total of 4 mils. For painted areas, paint with the paint used for shop painting. Apply paint by brush or spray to provide a minimum dry-film thickness of 2-mils.

-- End of Section --